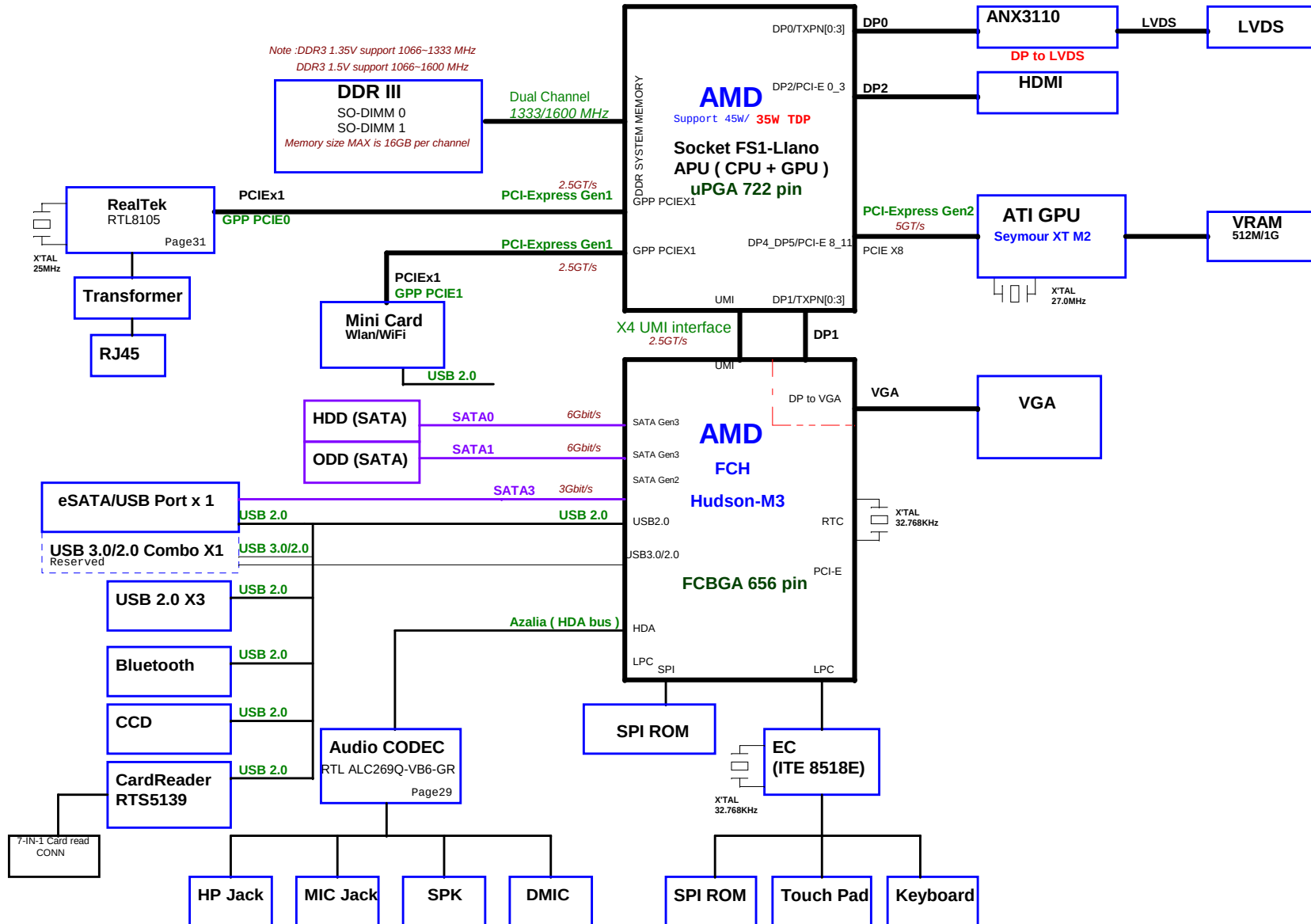


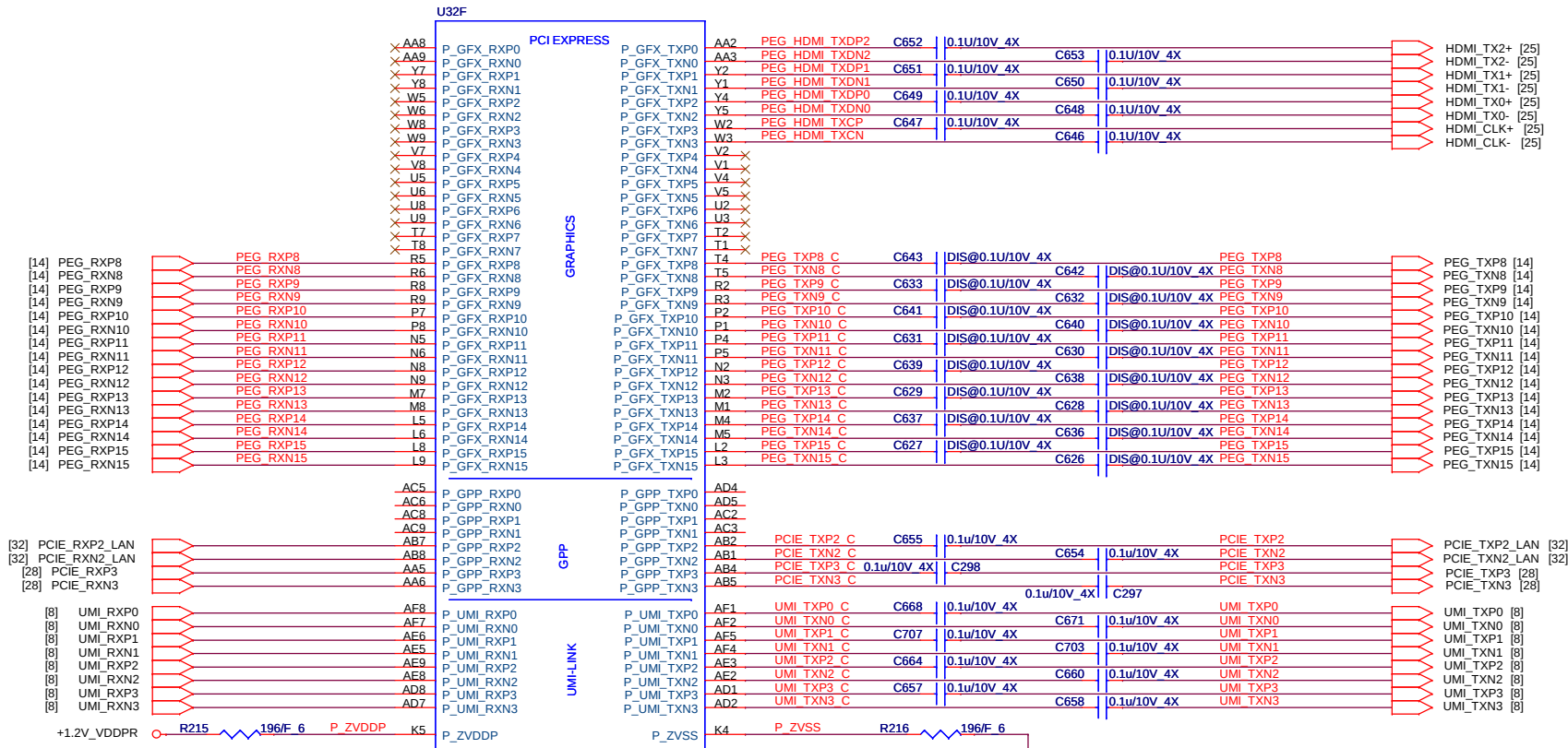
FAN /THERMAL
EMC2103-2
Page32

BLOCK DIAGRAM

Note :DP means Display Port Interface



Discharge	Page37
Charge	Page38
DDR3/0.75V (RT8207)	Page39
3V/5V (RT8206)	Page40
1.1V/+1.0V_GPU	Page41
+1.2V_VDDPR/+2.5	Page42
VDDI+VDDNB_CORE	Page43
GFX_CORE (0Z8117)	Page44
+1.8V_GPU (HPA00835RTER)	Page45

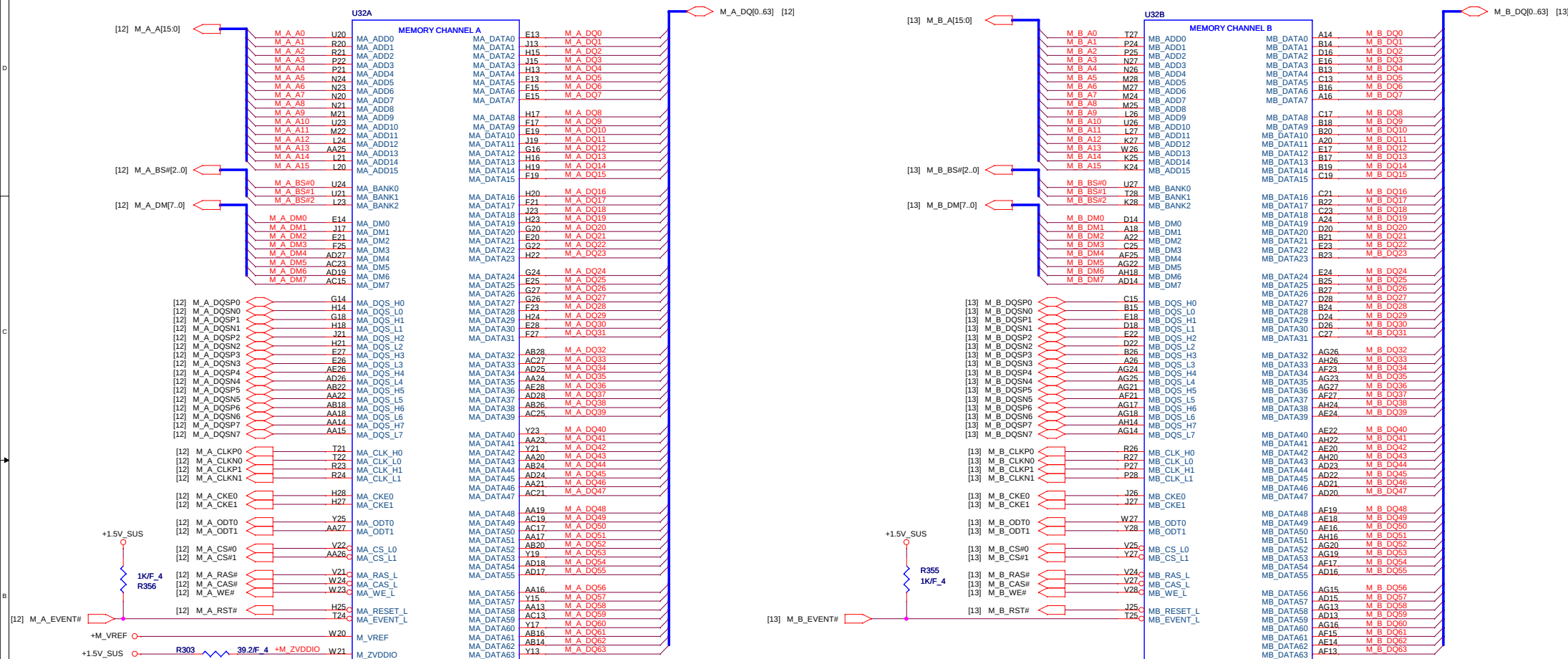


HDMI

P_GFX_TXP[N](3:0)
correspond to DisplayPort 2.

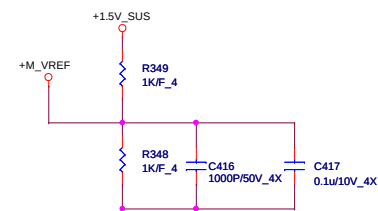
LAN
WLAN

Llano APU

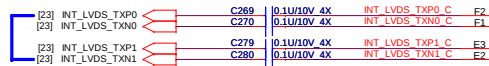
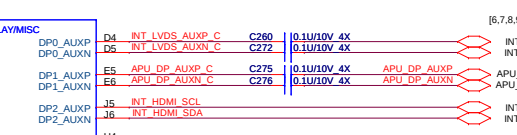


Liano APU

Liano APU

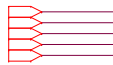


DP0 to LVDS

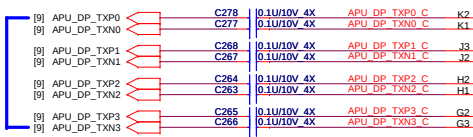
LVDS
VGA
HDMI

[6,7,8,9,10,11,12,13,19,23,24,25,26,27,28,29,30,31,32,33,35,36,37,38,39,40,42,43,44,45,46]
[9,19,23,25,26,29,30,33,34,35,36,37,38,39,44]
[8,10,24,29,32,34,35,36,38,39,40,41,43,46]
[5,5,6,9,10,12,13,37,38,40,42]
[6,8,9,23,25,28,29,42]

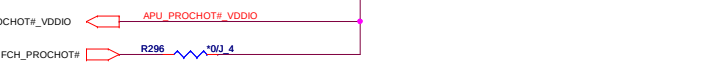
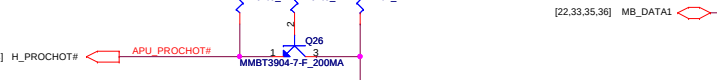
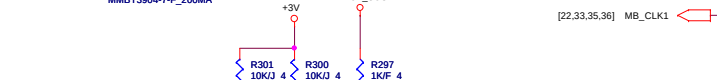
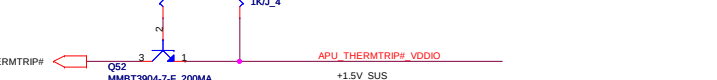
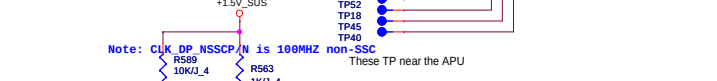
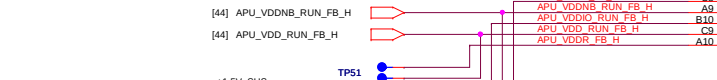
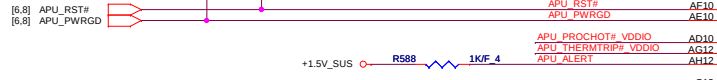
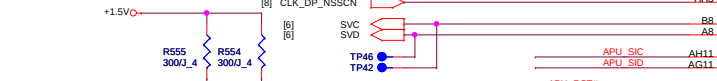
+3V
+5V
+3VPCU
+1.5V_SUS
+1.5V



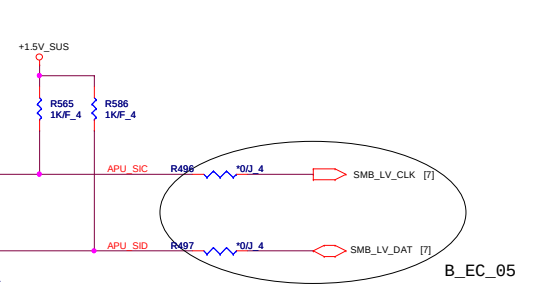
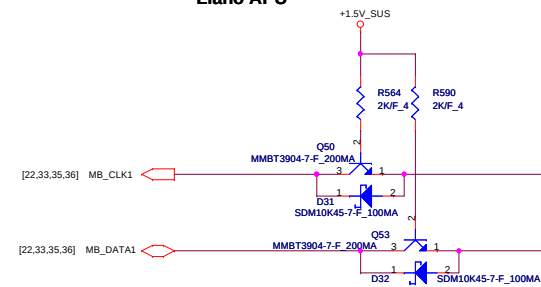
DP1 to Hudson-M3 VGA output



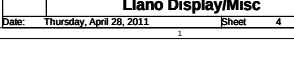
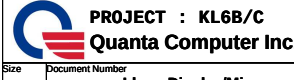
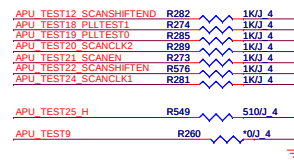
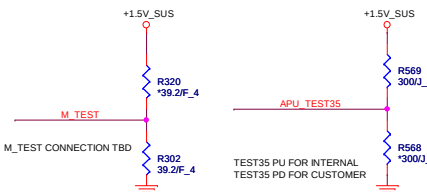
Note: CLK_APU_HCLKP/N is 100MHZ SSC



Llano APU



B_EC_05



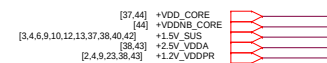
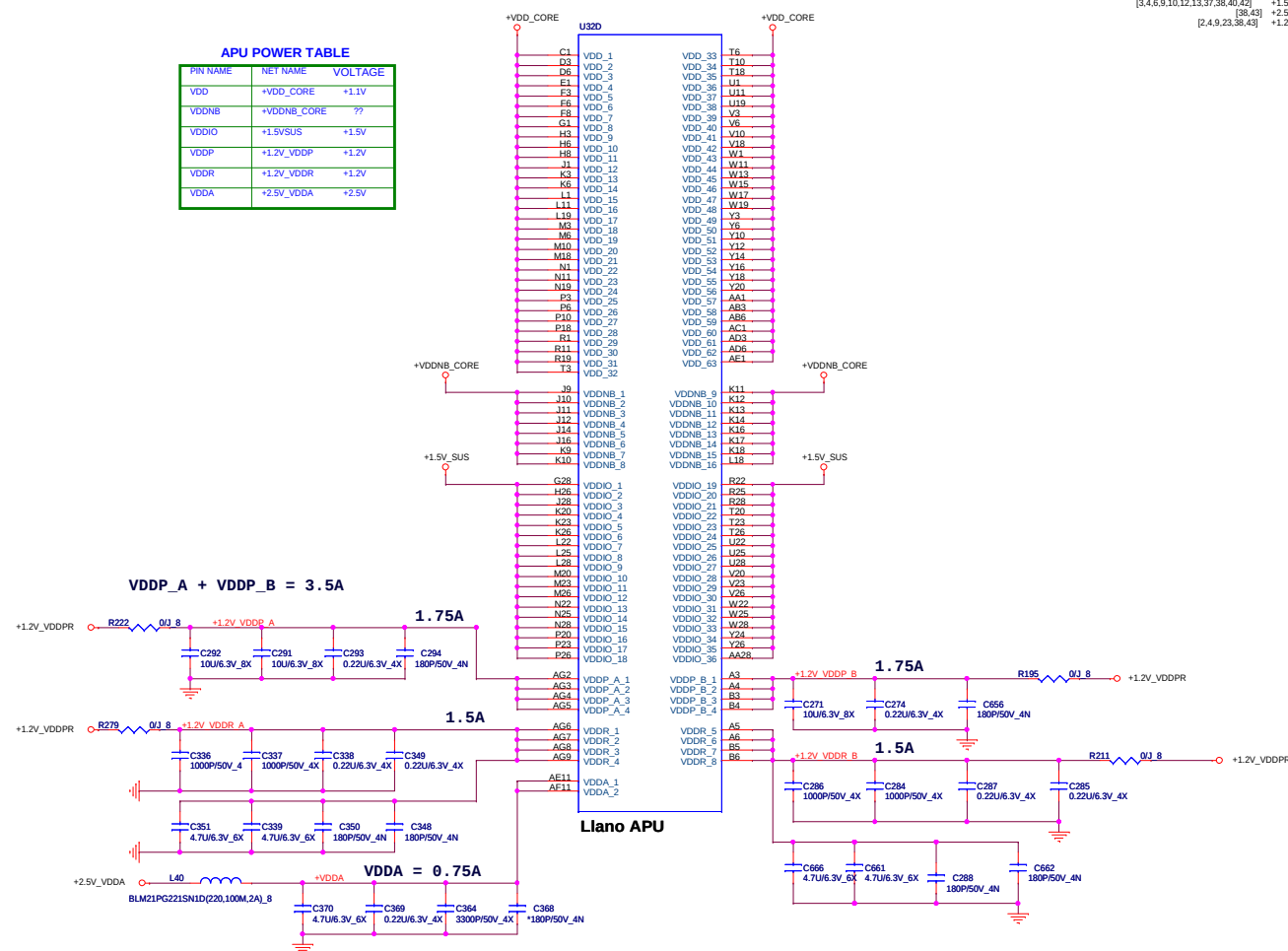
PROJECT : KL6B/C
Quanta Computer Inc.

Size Document Number
Llano Display/Misc
Date: Thursday, April 28, 2011 Sheet 4 of 49

Rev 1A

APU POWER TABLE

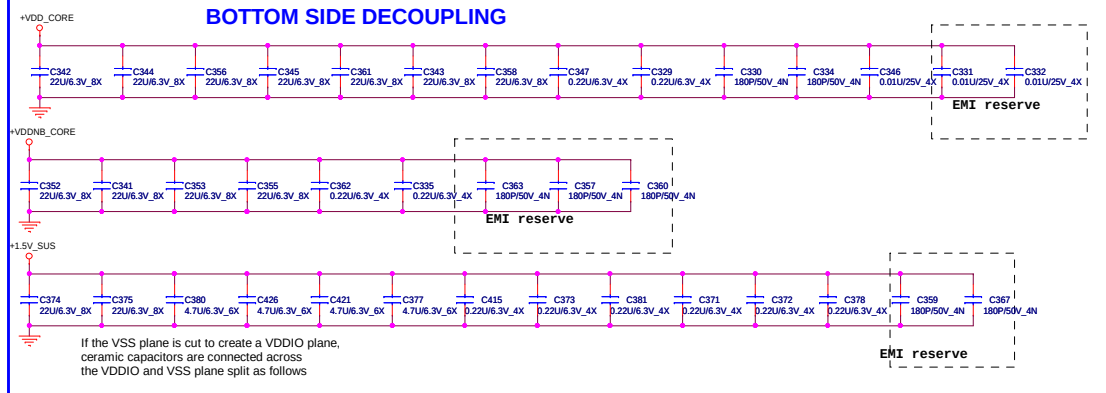
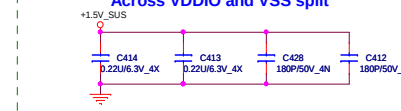
PIN NAME	NET NAME	VOLTAGE
VDD	+VDD_CORE	+1.1V
VDDNB	+VDDNB_CORE	??
VDDIO	+1.5VSUS	+1.5V
VDDP	+1.2V_VDDP	+1.2V
VDDR	+1.2V_VDDR	+1.2V
VDDA	+2.5V_VDDA	+2.5V



U32E		
A7	VSS_1	VSS_75
A13	VSS_2	VSS_76
A15	VSS_3	VSS_77
A17	VSS_4	VSS_78
A19	VSS_5	VSS_79
A21	VSS_6	VSS_80
A23	VSS_7	VSS_81
A25	VSS_8	VSS_82
B7	VSS_9	VSS_83
B9	VSS_10	VSS_84
C4	VSS_11	VSS_85
C10	VSS_12	VSS_86
C14	VSS_13	VSS_87
C16	VSS_14	VSS_88
C18	VSS_15	VSS_89
C20	VSS_16	VSS_90
C22	VSS_17	VSS_91
C24	VSS_18	VSS_92
C26	VSS_19	VSS_93
C28	VSS_20	VSS_94
D13	VSS_21	VSS_95
D15	VSS_22	VSS_96
D19	VSS_23	VSS_97
D21	VSS_24	VSS_98
D23	VSS_25	VSS_99
D25	VSS_26	VSS_100
D27	VSS_27	VSS_101
E4	VSS_28	VSS_102
E10	VSS_29	VSS_103
E12	VSS_30	VSS_104
F9	VSS_31	VSS_105
F11	VSS_32	VSS_106
F14	VSS_33	VSS_107
F16	VSS_34	VSS_108
F18	VSS_35	VSS_109
F20	VSS_36	VSS_110
F22	VSS_37	VSS_111
F24	VSS_38	VSS_112
F26	VSS_39	VSS_113
F28	VSS_40	VSS_114
G4	VSS_41	VSS_115
G8	VSS_42	VSS_116
G13	VSS_43	VSS_117
G15	VSS_44	VSS_118
G17	VSS_45	VSS_119
G19	VSS_46	VSS_120
G21	VSS_47	VSS_121
G23	VSS_48	VSS_122
G25	VSS_49	VSS_123
J4	VSS_50	VSS_124
J18	VSS_51	VSS_125
J20	VSS_52	VSS_126
J22	VSS_53	VSS_127
J24	VSS_54	VSS_128
J26	VSS_55	VSS_129
K19	VSS_56	VSS_130
L4	VSS_57	VSS_131
L7	VSS_58	VSS_132
L10	VSS_59	VSS_133
M0	VSS_60	VSS_134
M11	VSS_61	VSS_135
M19	VSS_62	VSS_136
N4	VSS_63	VSS_137
N7	VSS_64	VSS_138
N10	VSS_65	VSS_139
N18	VSS_66	VSS_140
P9	VSS_67	VSS_141
P11	VSS_68	VSS_142
P19	VSS_69	VSS_143
R4	VSS_70	VSS_144
R7	VSS_71	VSS_145
R10	VSS_72	VSS_146
R18	VSS_73	VSS_147
T9	VSS_74	VSS_148
		VSS_149

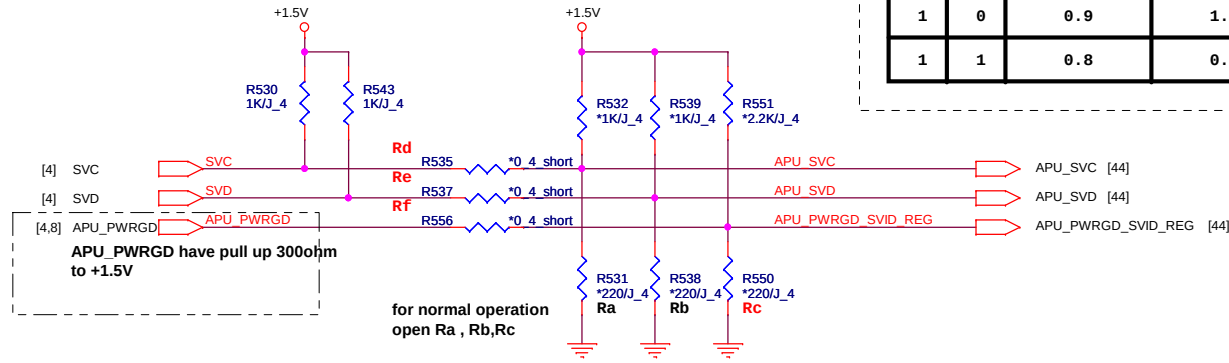
Llano APU

BOTTOM SIDE DECOUPLING

DECOUPLING between PROCESSOR and DIMMs
Across VDDIO and VSS split

VID Override Circuit

Note:
To override VID, Remove Rd, Re, Rf, install Rc
set VID via SVC & SVD option RES.



		BOOT VOLTAGE	
SVC	SVD	VFIX_+VDD =VCC/GND	VFIX_+VDD =OPEN
0	0	1.1	1.1
0	1	1.0	1.2
1	0	0.9	1.0
1	1	0.8	0.8

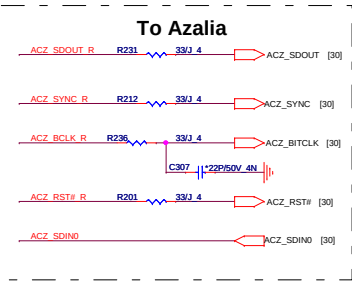
[4,8,9,23,25,28,29,42] +1.5V
[3,4,5,9,10,12,13,37,38,40,42] +1.5V_SUS



06

HDT+ Connector

Debug only

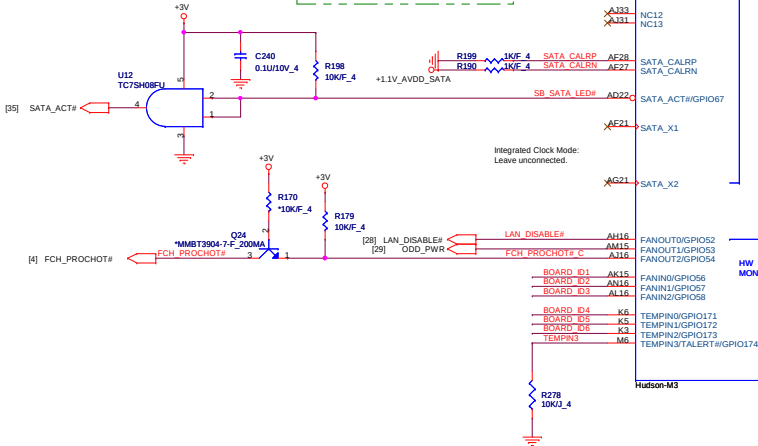


PLACE SATA AC COUPLING
CAPS CLOSE TO HUDSON-M2/M3

SATA HDD/SSD

SATA ODD

ESATA#1

SATA PORTS DISTRIBUTION:
0. 2.5 INCH DISK DRIVER
1. SATA ODD
2. 2.5 INCH DISK DRIVER
3. eSATA
4-7. NOT USEDPLACE SATA_CAL RES VERY
CLOSE TO BALL OF
HUDSON-M2/M3

HUDSON-M2

Part 2 of 5

SD CARD

LAN

SERIAL ATA

ROM

VGA DAC

VGA MAINLINK

HW MONITOR

Hudson-M3

Part 2 of 5

SD CARD

LAN

SERIAL ATA

ROM

VGA DAC

VGA MAINLINK

HW MONITOR

Hudson-M3

Part 2 of 5

SD CARD

LAN

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Hudson-M3

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VGA MAINLINK

HW MONITOR

Hudson-M3

Part 2 of 5

SD CARD

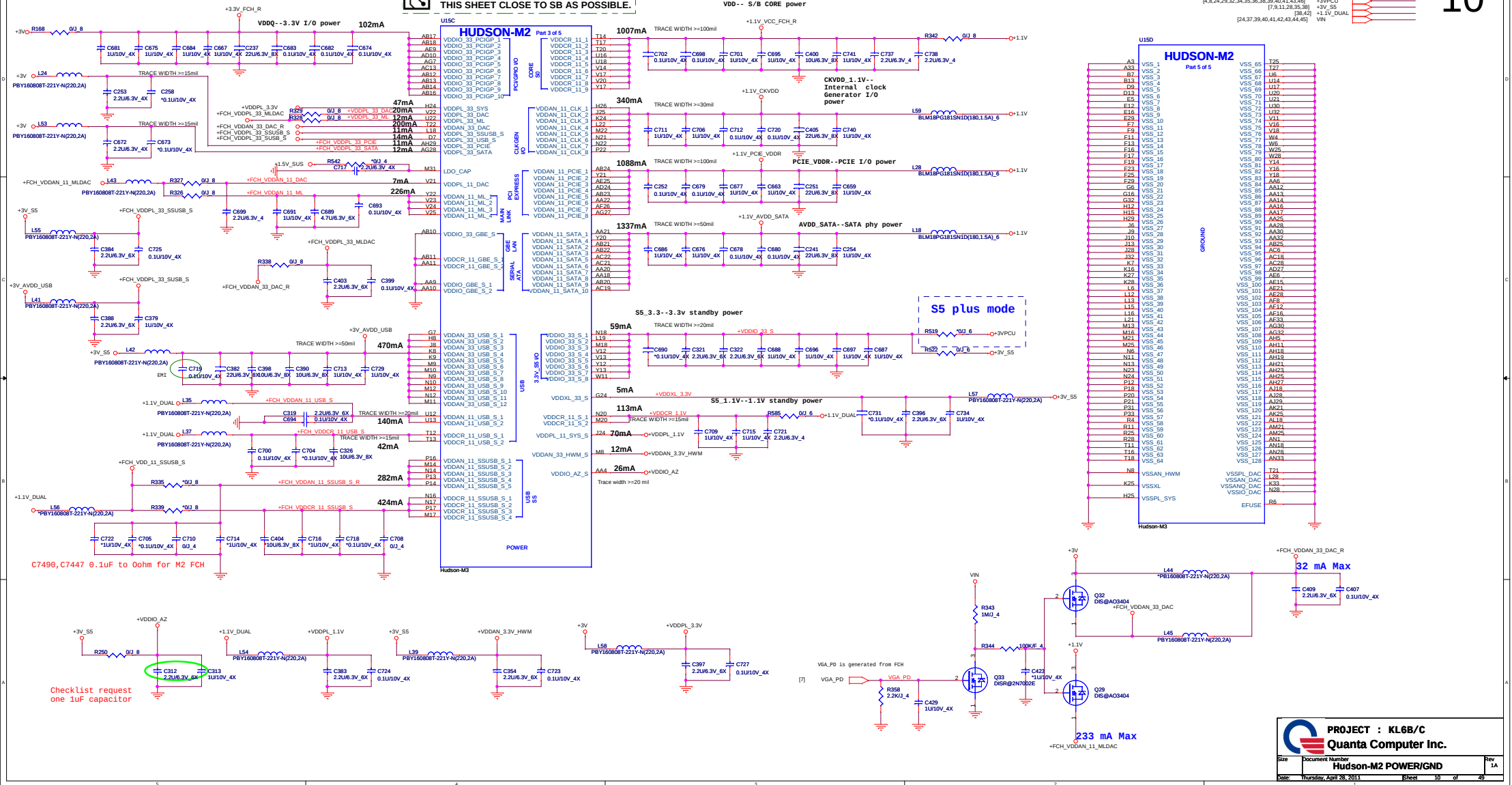
LAN

PLACE ALL THE DECOUPLING CAPS ON THIS SHEET CLOSE TO SB AS POSSIBLE.

VDD - - S/B CORE power

[4,6,7,8,9,11,12,13,19,23,24,25,26,27,28,29,30,31,32,33,35,36,37,38,39,40,42,43,44,45,46]
[4,8,24,29,32,34,35,36,38,39,40,41,43,46]
[7,9,11,28,35,38]
[24,37,39,40,41,42,43,44,46]
[38]

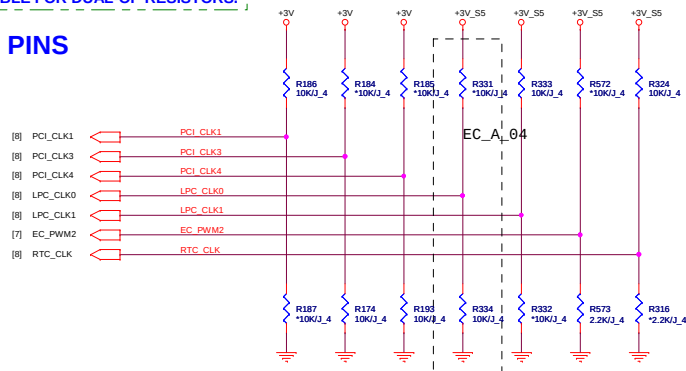
	+3V
[42]	+1.1V
[46]	+3VPCU
[5,38]	+3V_S5
[38,42]	+1.1V_DU
[4,45]	VIN





OVERLAP COMMON PADS WHERE POSSIBLE FOR DUAL-OP RESISTORS.

STRAPS PINS

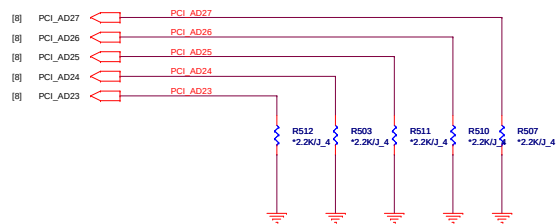


REQUIRED STRAPS

		PCI_CLK1		PCI_CLK3	PCI_CLK4	LPC_CLK0	LPC_CLK1	EC_PWM2	RTC_CLK
PULL HIGH	*****	ALLOW PCIE Gen2 DEFAULT	*****	USE DEBUG STRAP	non_Fusion CLOCK MODE	EC ENABLED	CLKGEN ENABLED DEFAULT	LPC ROM	S5 PLUS MODE DISABLED DEFAULT
PULL LOW	*****	FORCE PCIE Gen1	*****	IGNORE DEBUG STRAP DEFAULT	FUSION CLOCK MODE DEFAULT	EC DISABLED DEFAULT	CLKGEN DISABLED	SPI ROM DEFAULT	S5 PLUS MODE ENABLED

DEBUG STRAPS

FCH HAS 15K INTERNAL PU FOR PCI_AD[27:23]

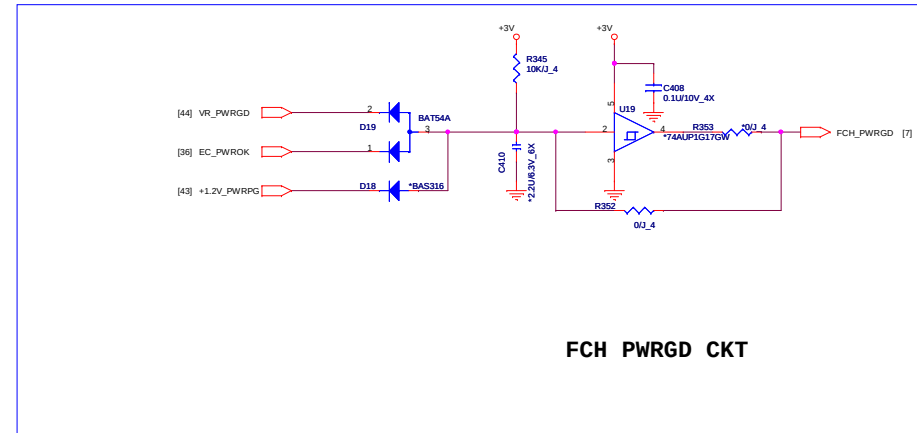


	PCI_AD27	PCI_AD26	PCI_AD25	PCI_AD24	PCI_AD23
PULL HIGH	USE PCI PLL DEFAULT	DISABLE ILA AUTORUN DEFAULT	USE FC PLL DEFAULT	USE DEFAULT PCIE STRAPS DEFAULT	DISABLE PCI MEM BOOT DEFAULT
PULL LOW	BYPASS PCI PLL	ENABLE ILA AUTORUN	BYPASS FC PLL	USE EEPROM PCIE STRAPS	ENABLE PCI MEM BOOT

[4,6,7,8,9,10,12,13,19,23,24,25,26,27,28,29,30,31,32,33,35,36,37,38,39,40,42,43,44,45,46] +3V
[7,9,10,28,35,38] +3V_S5



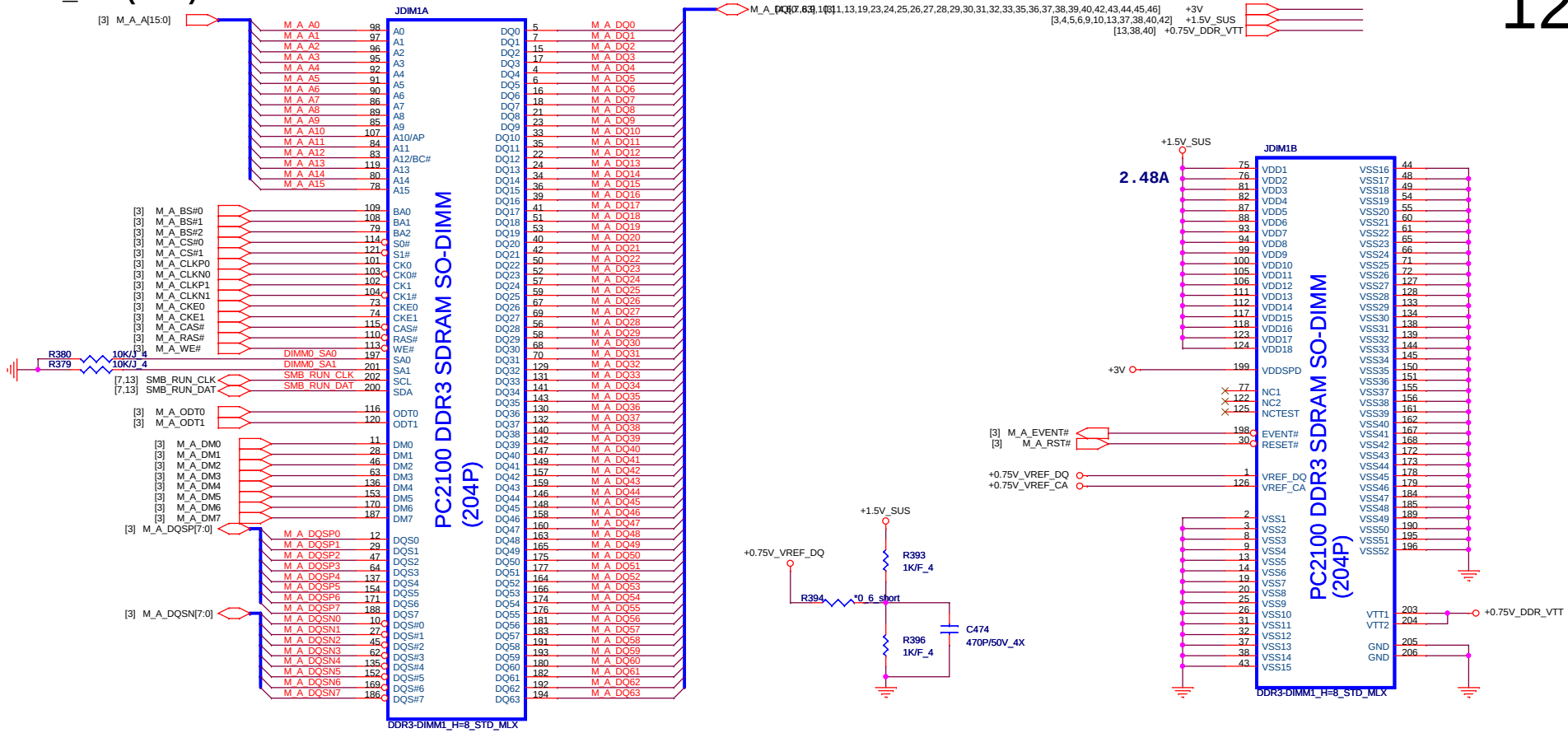
11



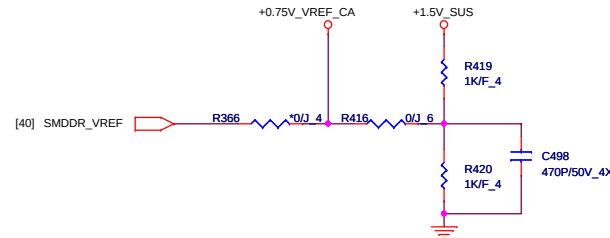
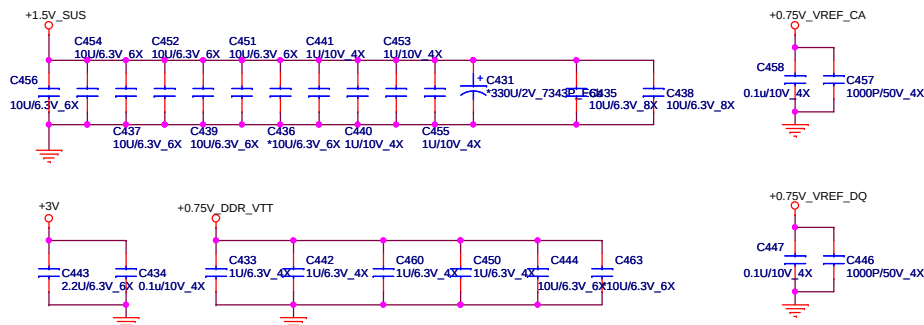
FCH_PWRGD CKT

DDR STD(DDR)

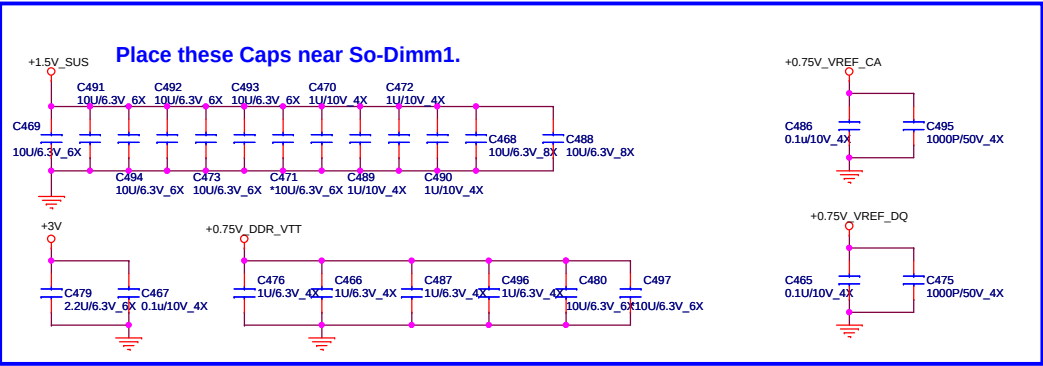
12



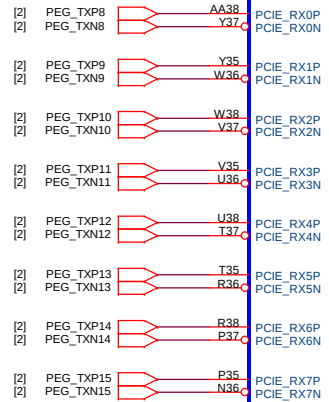
Place these Caps near So-Dimm0.



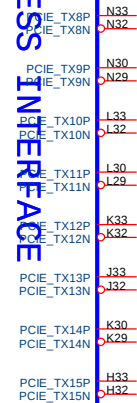
13



5GT/s bit rate



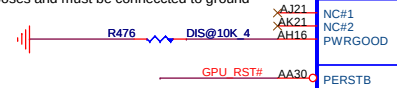
PCIE EXPRESS INTERFACE



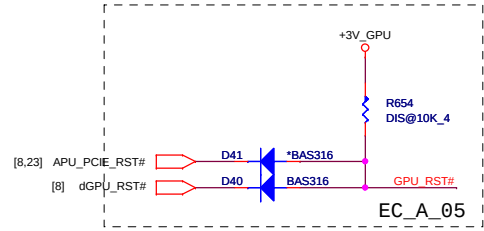
CALIBRATION

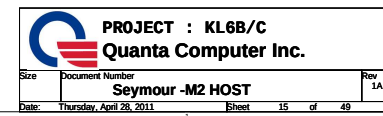


For M97 only Madison and Park the PWRGOOD ball is for test purposes and must be connected to ground

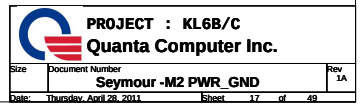


DIS@Seymour_M2





17

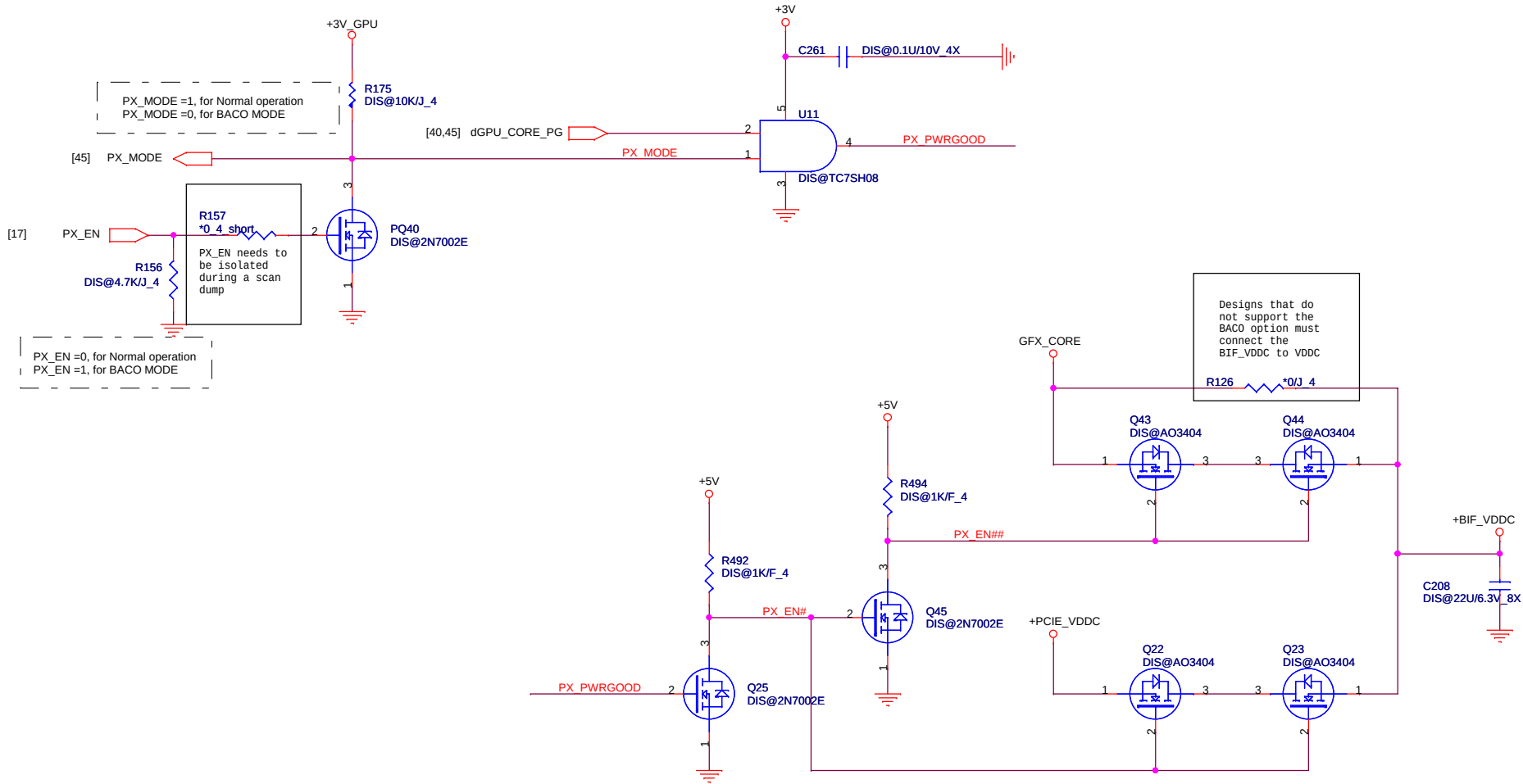




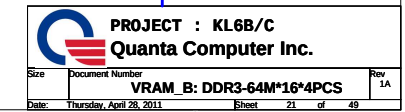
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 [9,23,25,26,29,30,33,34,35,36,37,38,39,44]
 [17]
 [14,15,16,17,22,38,40,46]
 [17,37,45]

+3V
 +5V
 +PCIE_VDDC
 +3V_GPU
 GFX_CORE

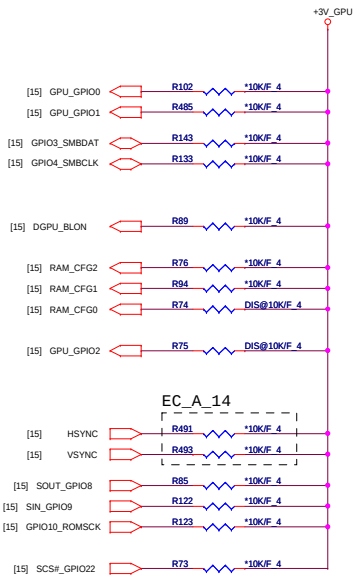
19



21



PIN STRAPS



Memory Aperture size

RAM_CFG[2:0] Size

000	128MB
001	256MB
010	64MB
011	32MB

ROM Table

HSYNC	VSYNC	Discription
0	0	No Audio
0	1	Any one by dectec
1	0	DP only
1	1	Both DP & HDMI

CONFIGURATION STRAPS

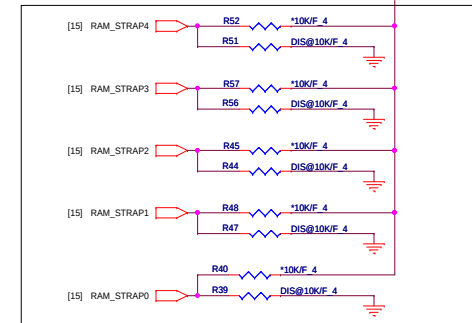
STRAPS	PIN	DESCRIPTION OF DEFAULT SETTINGS	DEFAULT	REMARK
TX_PWRS_ENB	GPIO0	0 = 50% TX OUTPUT SWING 1 = FULL TX OUTPUT SWING	0	
TX_DEEMPH_EN	GPIO1	PCIE TRANSMITTER DE-EMPHASIS ENABLED 0 = TX DE-EMPHASIS DISABLED 1 = TX DE-EMPHASIS ENABLED	0	
BIOS_ROM_EN	GPIO_22_ROMCSB	ENABLE EXTERNAL BIOS ROM (Only for GDDR5) 0 = DISABLE 1 = ENABLE	0	
ROMIDCFG(2:0)	GPIO[13:11]	SERIAL ROM TYPE OR MEMORY APERTURE SIZE SELECT NUMONYX M2SP10A : 101	000	See ROM table
BIF_GEN2_EN_A	GPIO2	0 = PCIE DEVICE AS 2.5GT/S CAPABLE 1 = PCIE DEVICE AS 5GT/S CAPABLE	1	
GPIO_8_ROMSO H2SYNC GPIO_21_BB_EN	GPIO8 H2SYNC GPIO21	Reserved Only	0	
AUD[1] AUD[0]	HSYNC VSYNC	AUD[1:0] 00: NO AUDIO FUNCTION. 01: AUDIO FOR DISPLAYPORT AND HDMI IF ADAPTER IS DETECTED 10: AUDIO FOR DISPLAYPORT ONLY. 11: AUDIO FOR BOTH DISPLAYPORT AND HDMI.	11	See Audio table
GPIO_9_ROMSI	GPIO9	0 = VGA controller capacity enable	0	
VIP_DEVICE_STRAP_ENA VIP: Video Capture Port Interface	V2SYNC	0 = DRIVER would ignore the value sample on VHAD_0 during RESET.	0	

[15,17,18,38,46] +1.8V_GPU
[14,15,16,17,19,38,40,46] +3V_GPU

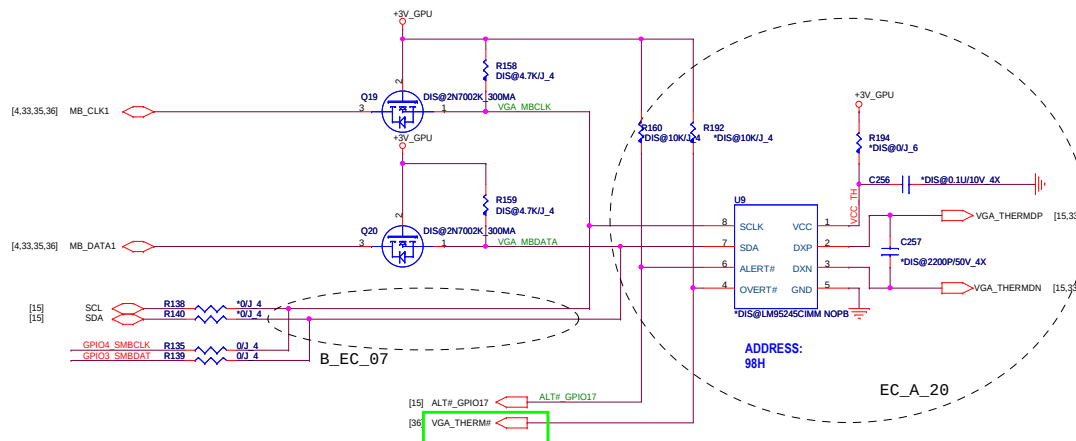
VRAM Memory TYPE

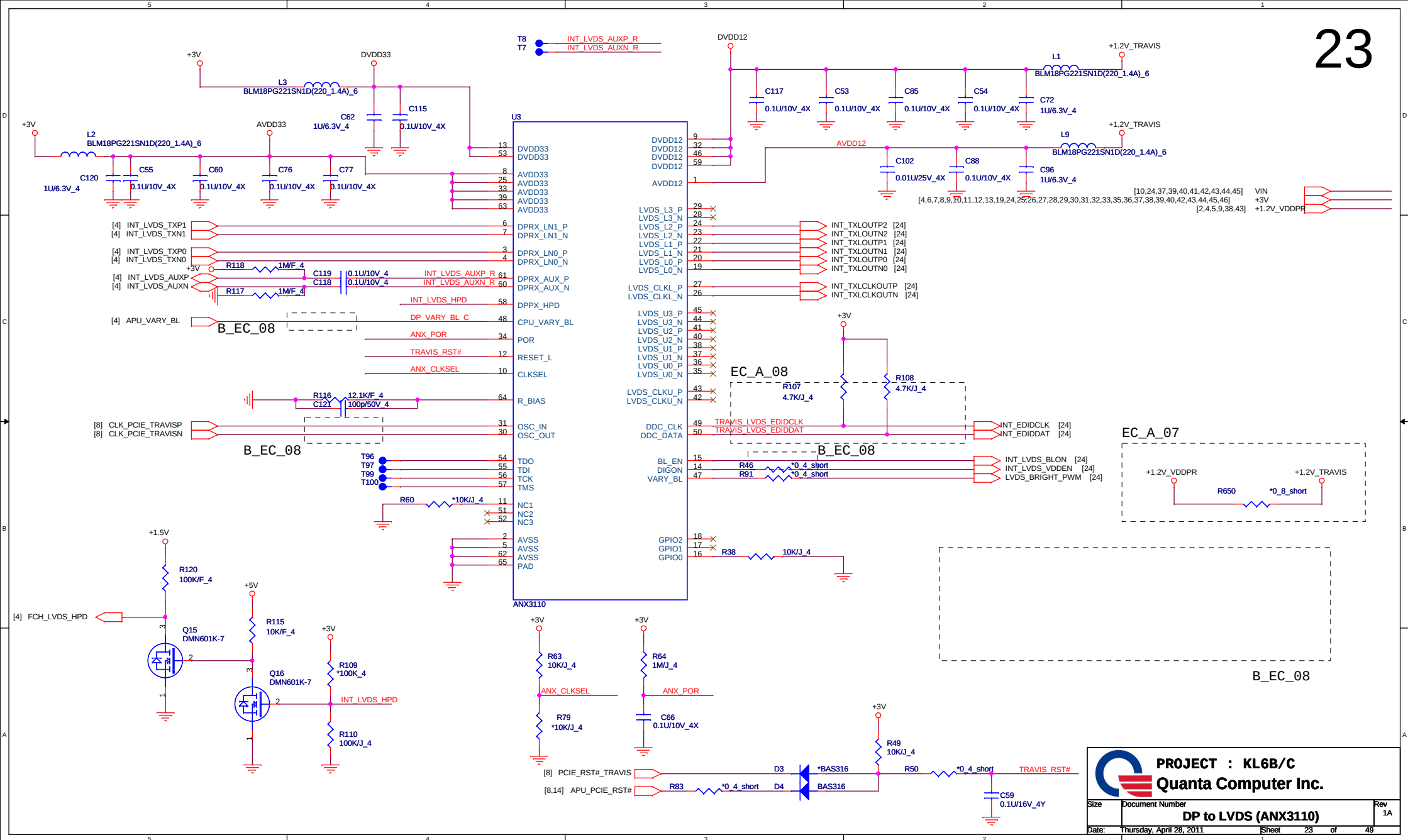
Vendor	Vendor P/N	STN B/S P/N	Size	RAM_STRAP3 DVPDATA_3	RAM_STRAP2 DVPDATA_2	RAM_STRAP1 DVPDATA_1	RAM_STRAP0 DVPDATA_0	RAM_STRAP4 15"	RAM_STRAP4 14"
Hynix	H5TQ1G63DFR-11C	AKD5LZWTW02 (64M*16-1Gb)	512MB	0	0	1	0	0	1
	H5TQ2G63BFR-11C	AKD5MGWTW00 (128M*16-1Gb)	1GB	0	0	0	0	0	1
Samsung	K4W1G1646G-BC11	AKD5EGGT500 (64M*16-1Gb)	512MB	0	0	1	1	0	1
	K4W2G1646C-HC11	AKD5MGWT500 (128M*16-1Gb)	1GB	0	0	0	1	0	1

Make sure to use the correct VRAM setting

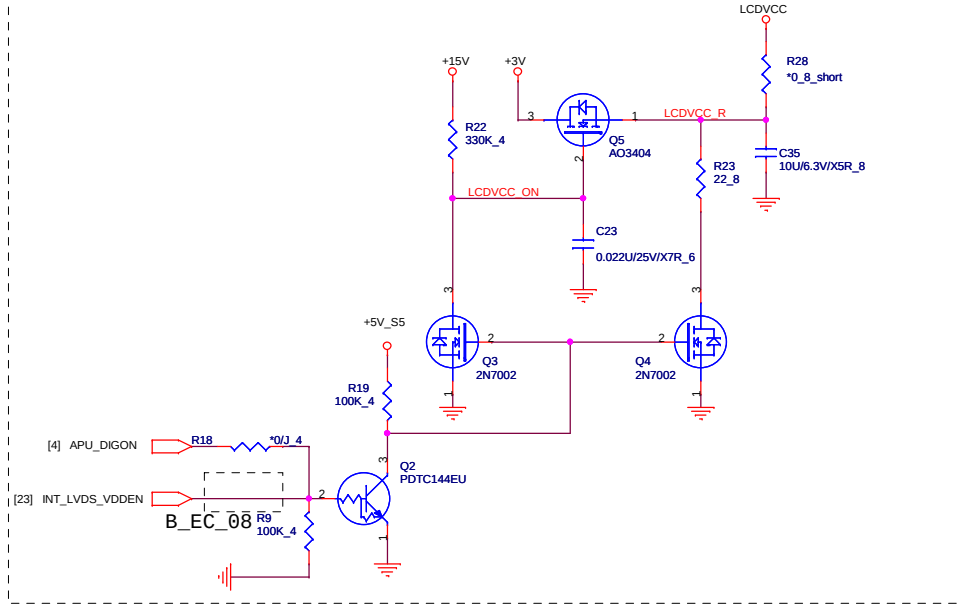


Thermal Sensor

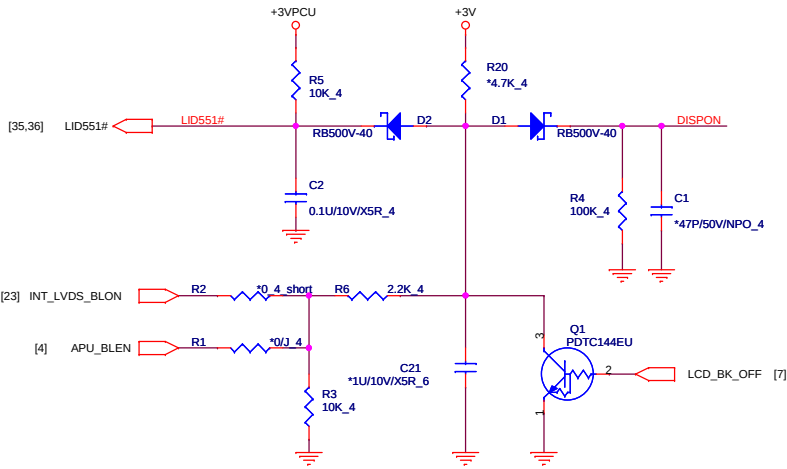




LCDVCC

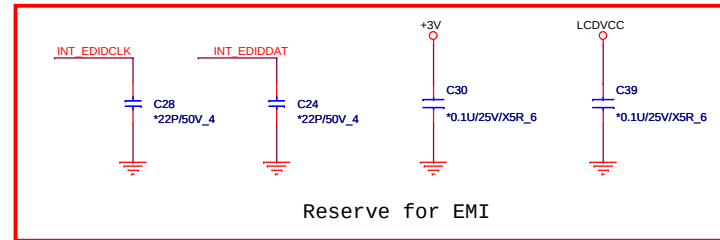
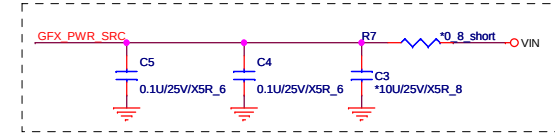
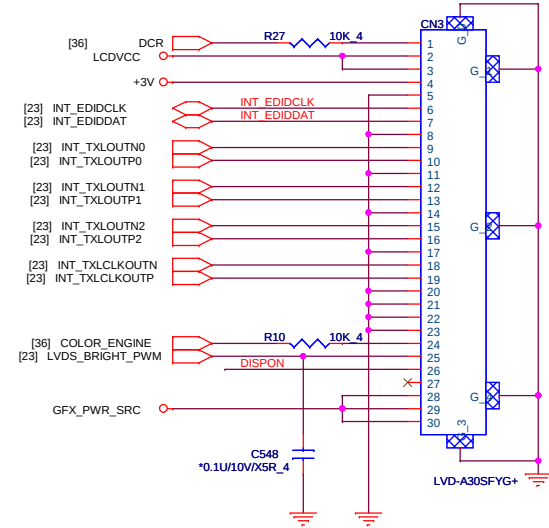


back light

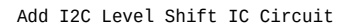


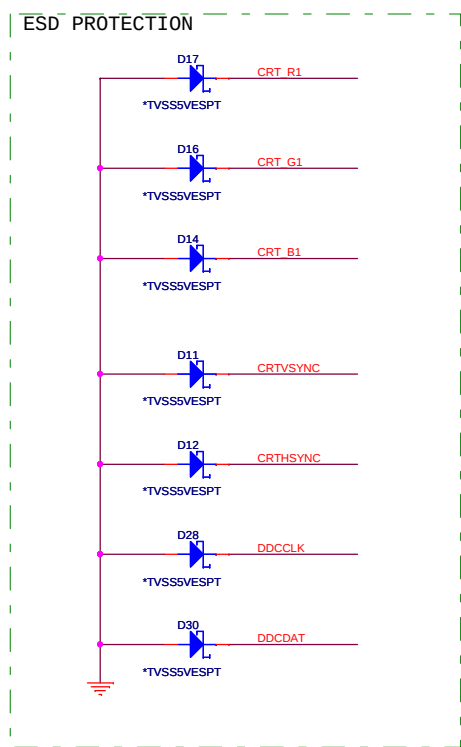
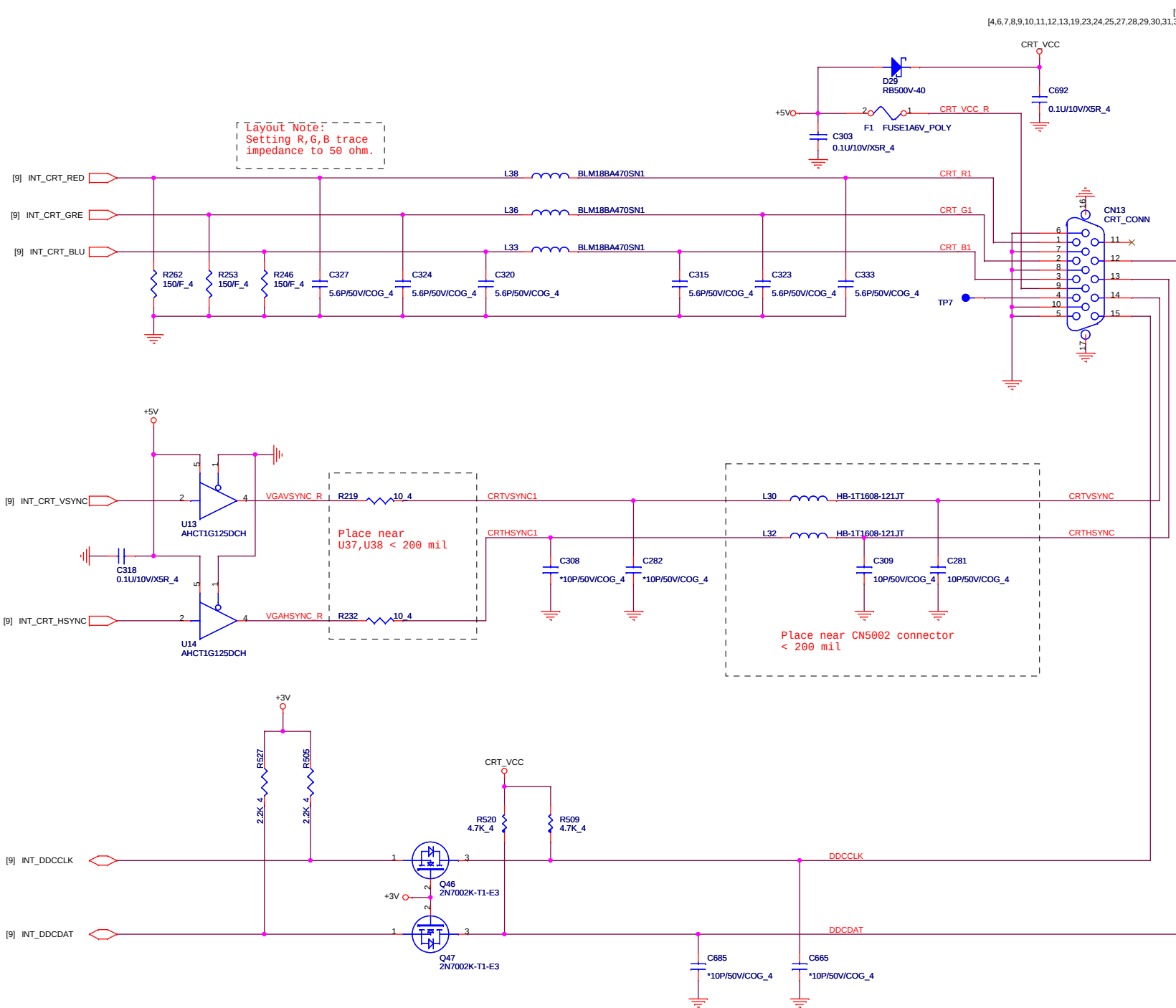
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[29,31,35,38,40,41,42,46]
[4,8,10,29,32,34,35,36,38,39,40,41,43,46]
[10,37,39,40,41,42,43,44,45]

+3V
+15V
+3VPCU
VIN



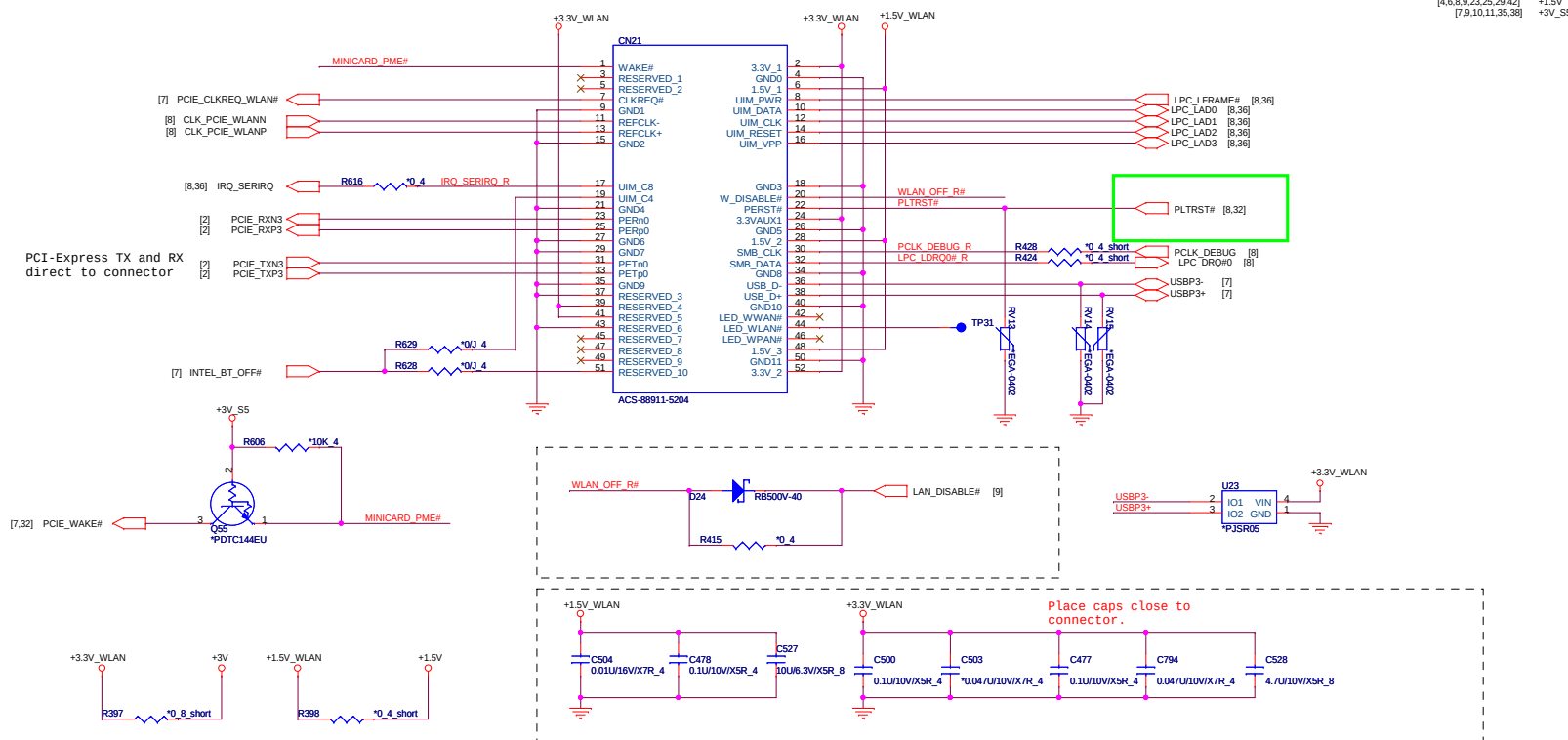
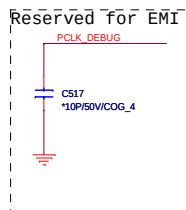
Reserve for EMI





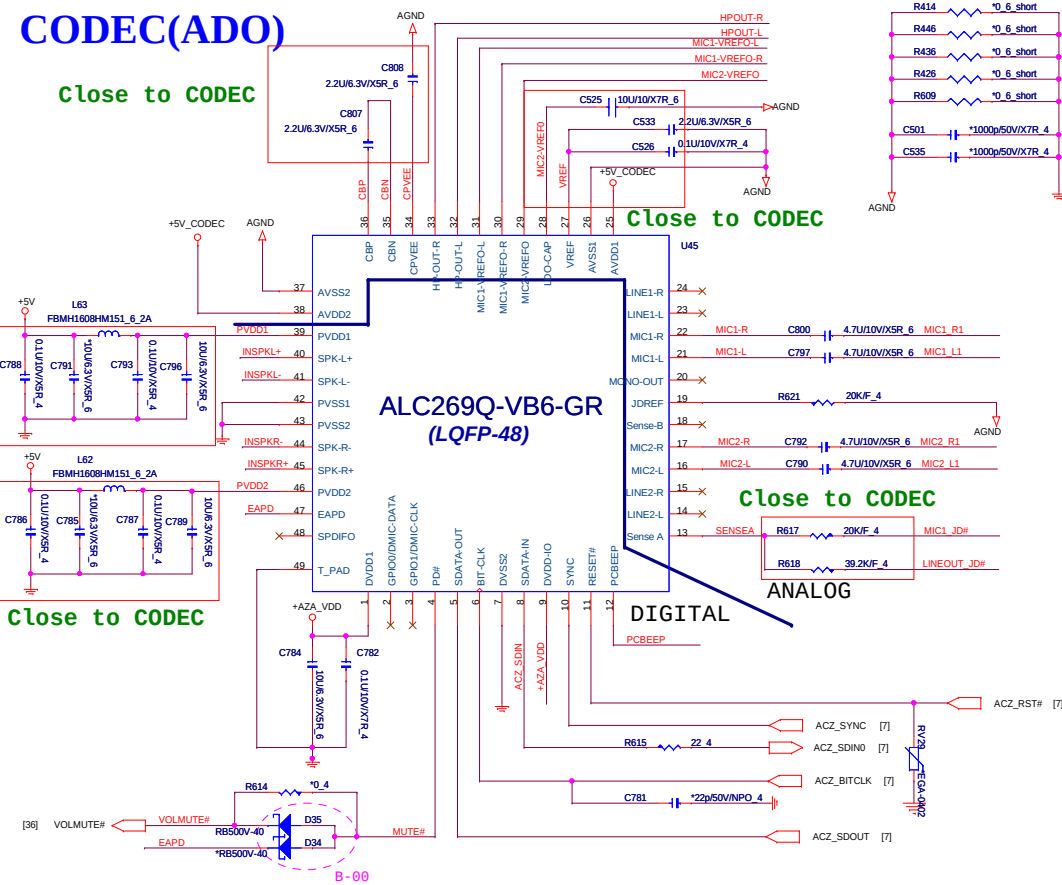
MiniCard WLAN connector

28



CODEC(ADO)

Close to CODEC



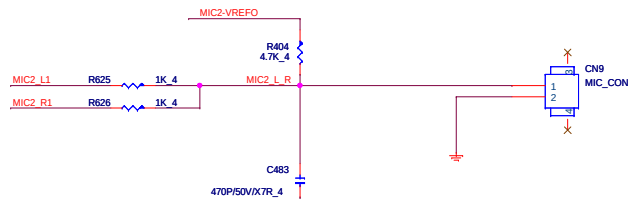
ALC269Q-VB6-GR
(LQFP-48)

Close to CODEC

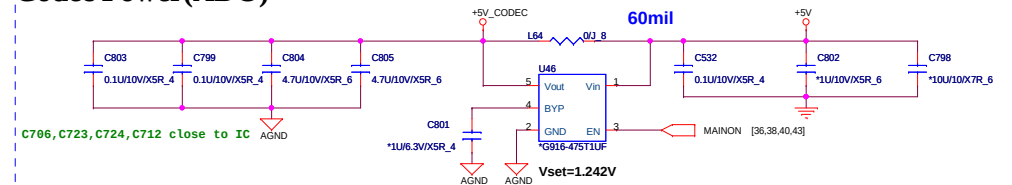
ANALOG

DIGITAL

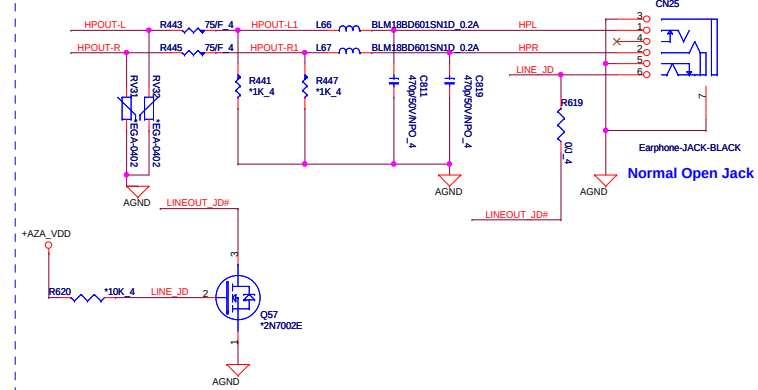
INTERNAL MIC



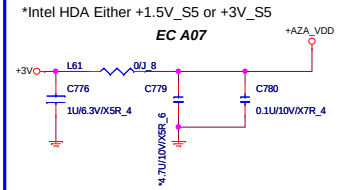
Codec Power(ADO)



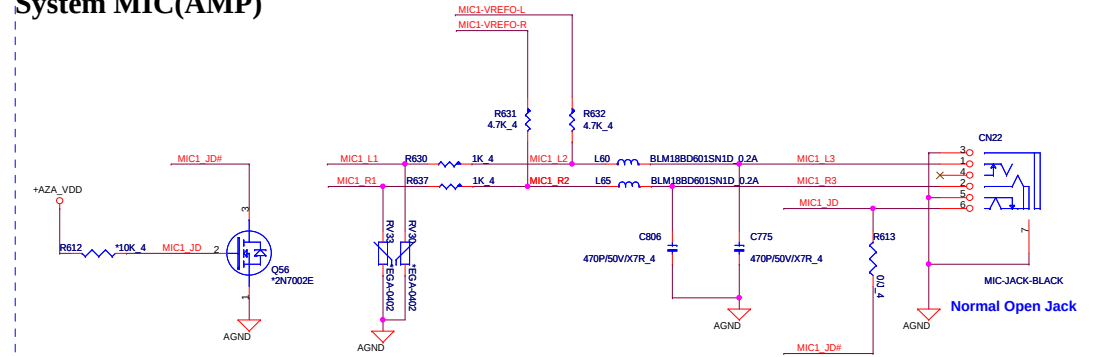
Earphone(AMP)



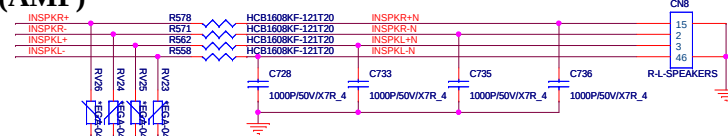
HDA Power(ADO)



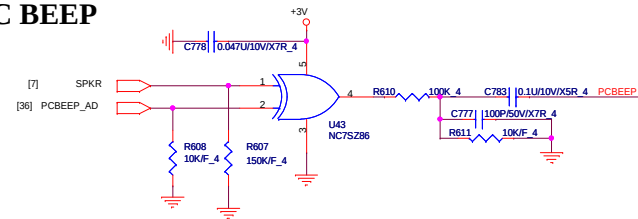
System MIC(AMP)



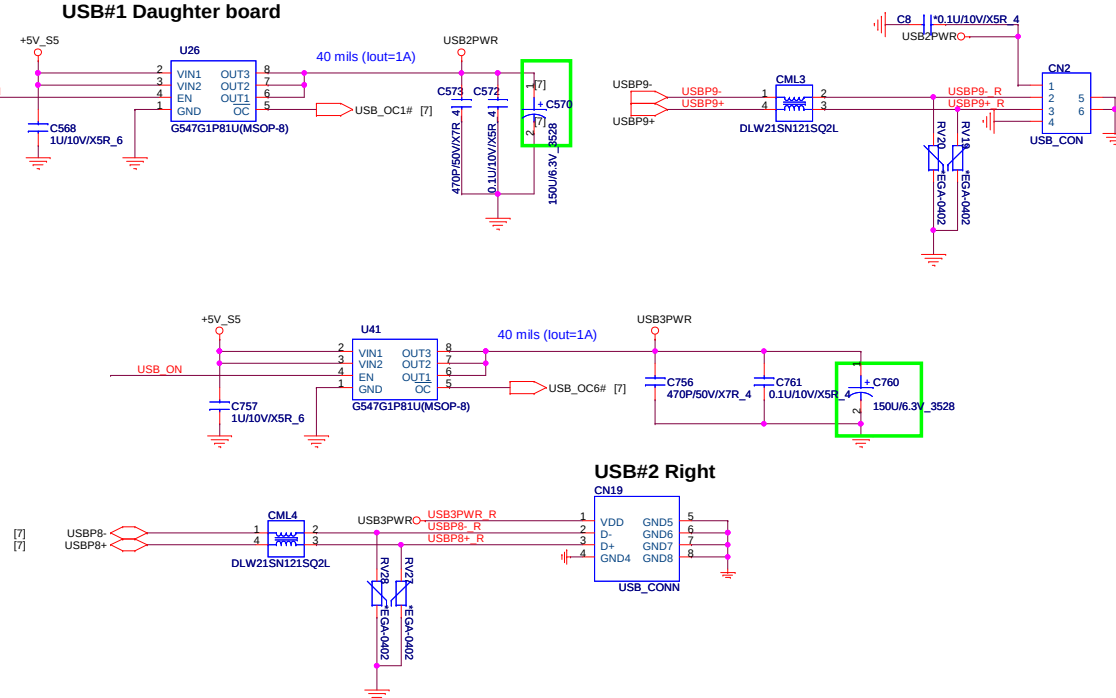
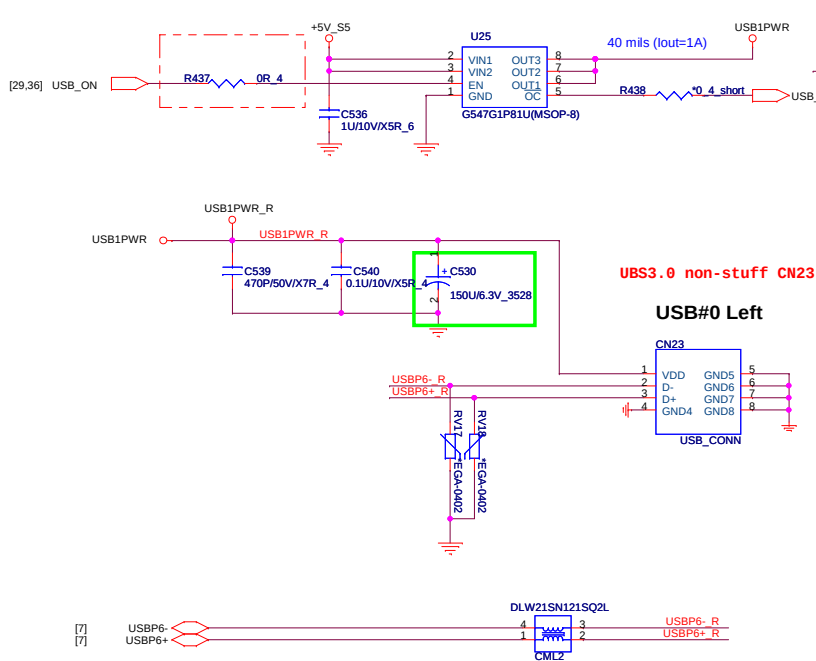
Speaker(AMP)



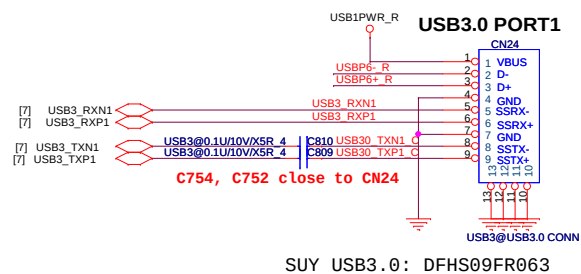
PC BEEP



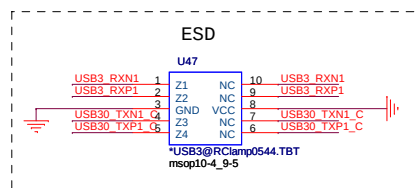
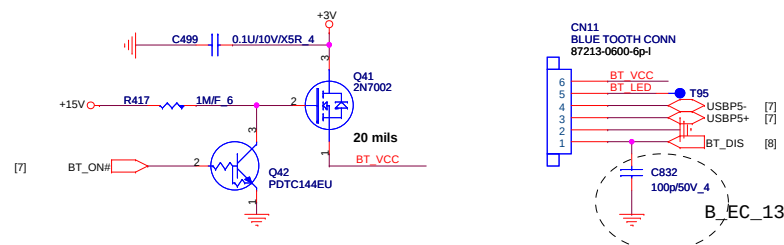
USB#1 Daughter board



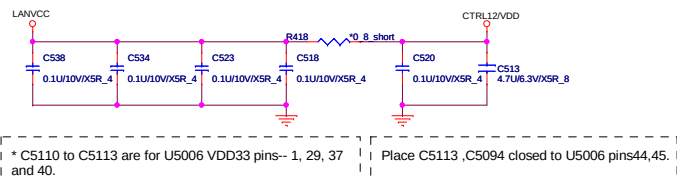
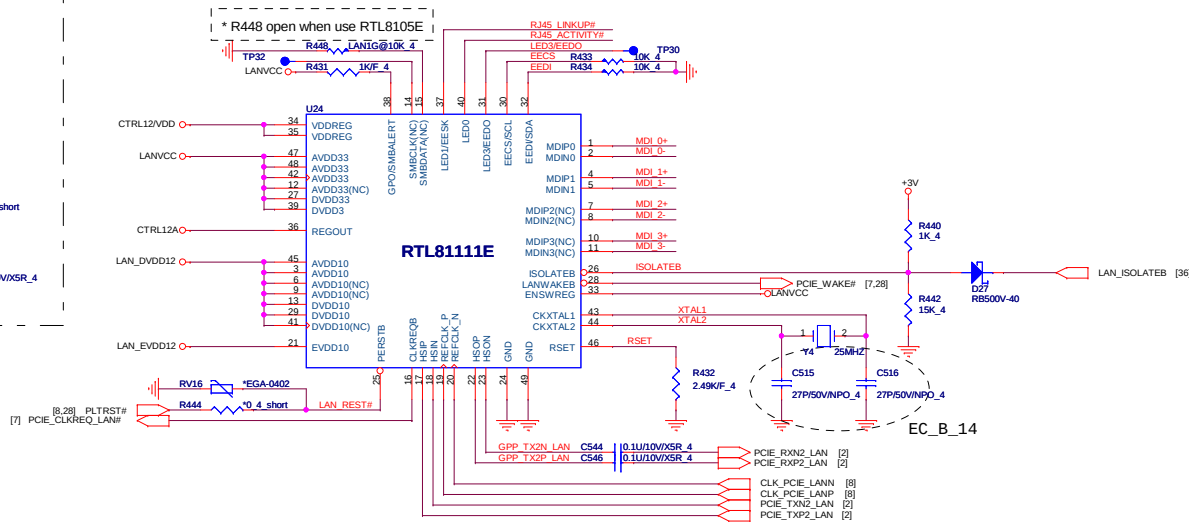
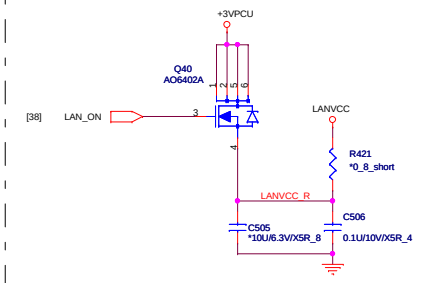
REAR USB PORT X1



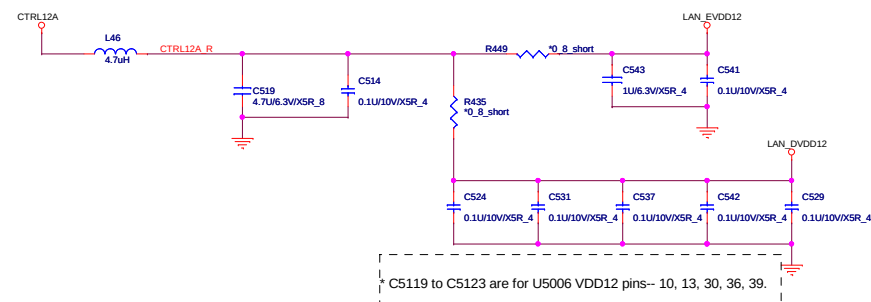
BLUETOOTH



LANVCC



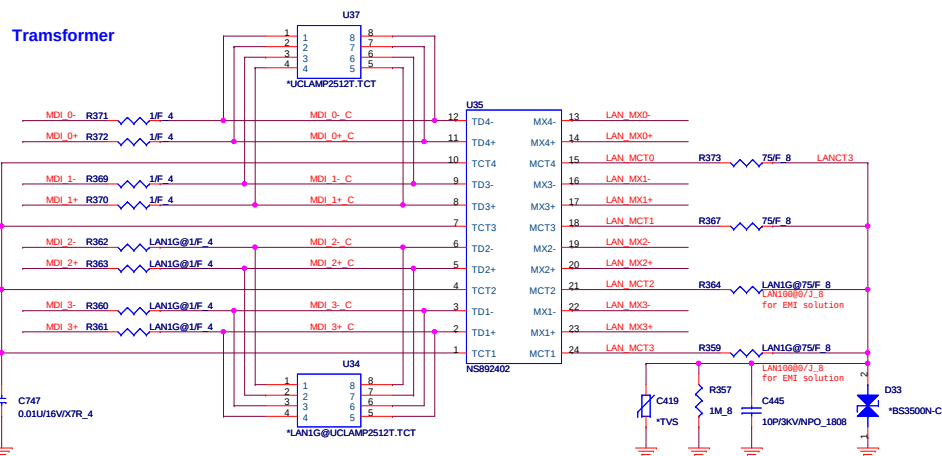
Place C5113, C5094 closed to U5006 pins 44, 45, 46 and 40.



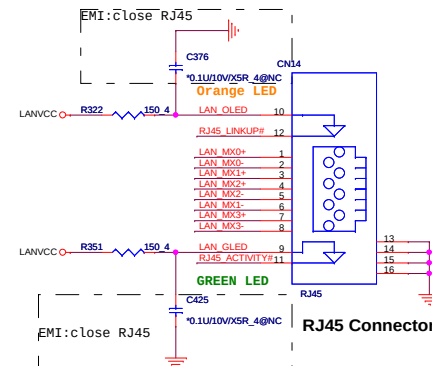
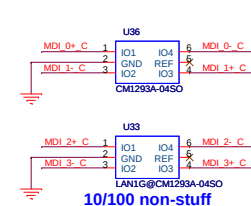
* C5119 to C5123 are for U5006 VDD12 pins-- 10, 13, 30, 36, 39.

Layout: All termination signal should have 20 mil trace

Transformer



RJ45 Connector



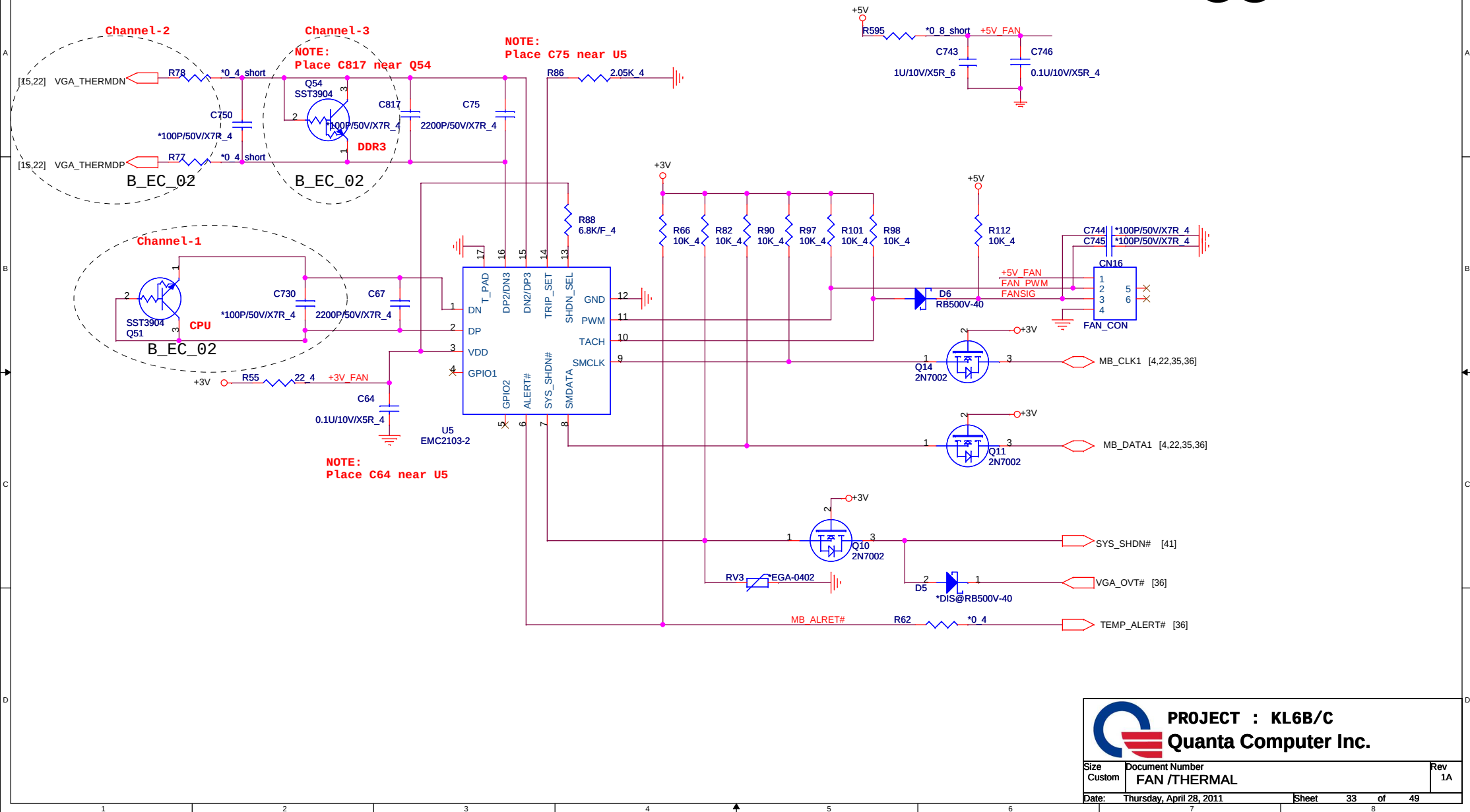
FAN CONTROL

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[9,19,23,25,26,29,30,34,35,36,37,38,39,44]

+3V
+5V

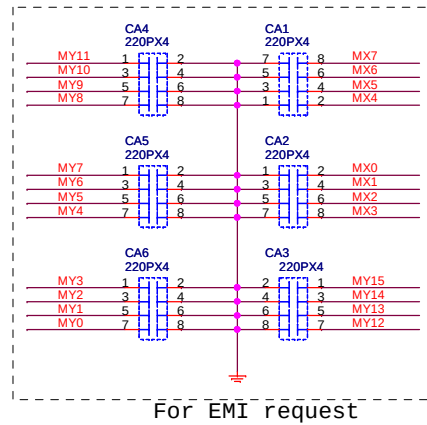
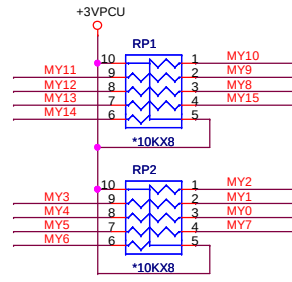
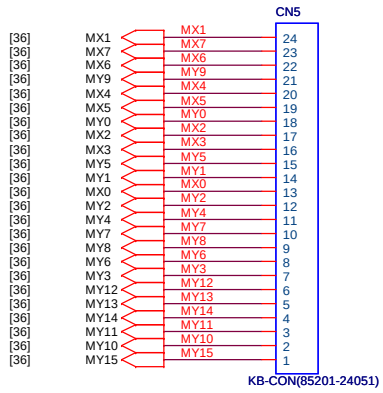


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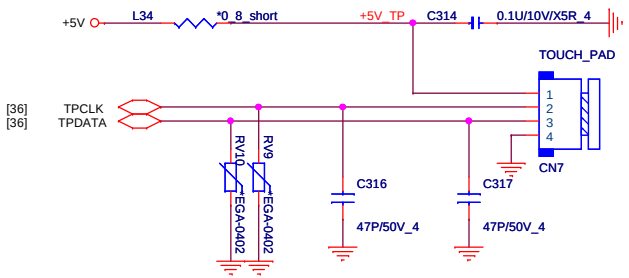


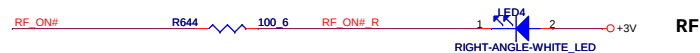
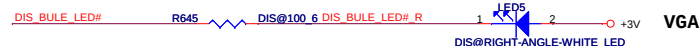
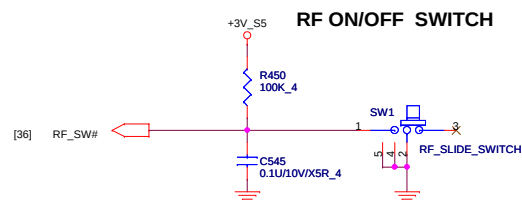
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Size Custom	Document Number FAN /THERMAL	Rev 1A
Date: Thursday, April 28, 2011	Sheet 33 of 49	7

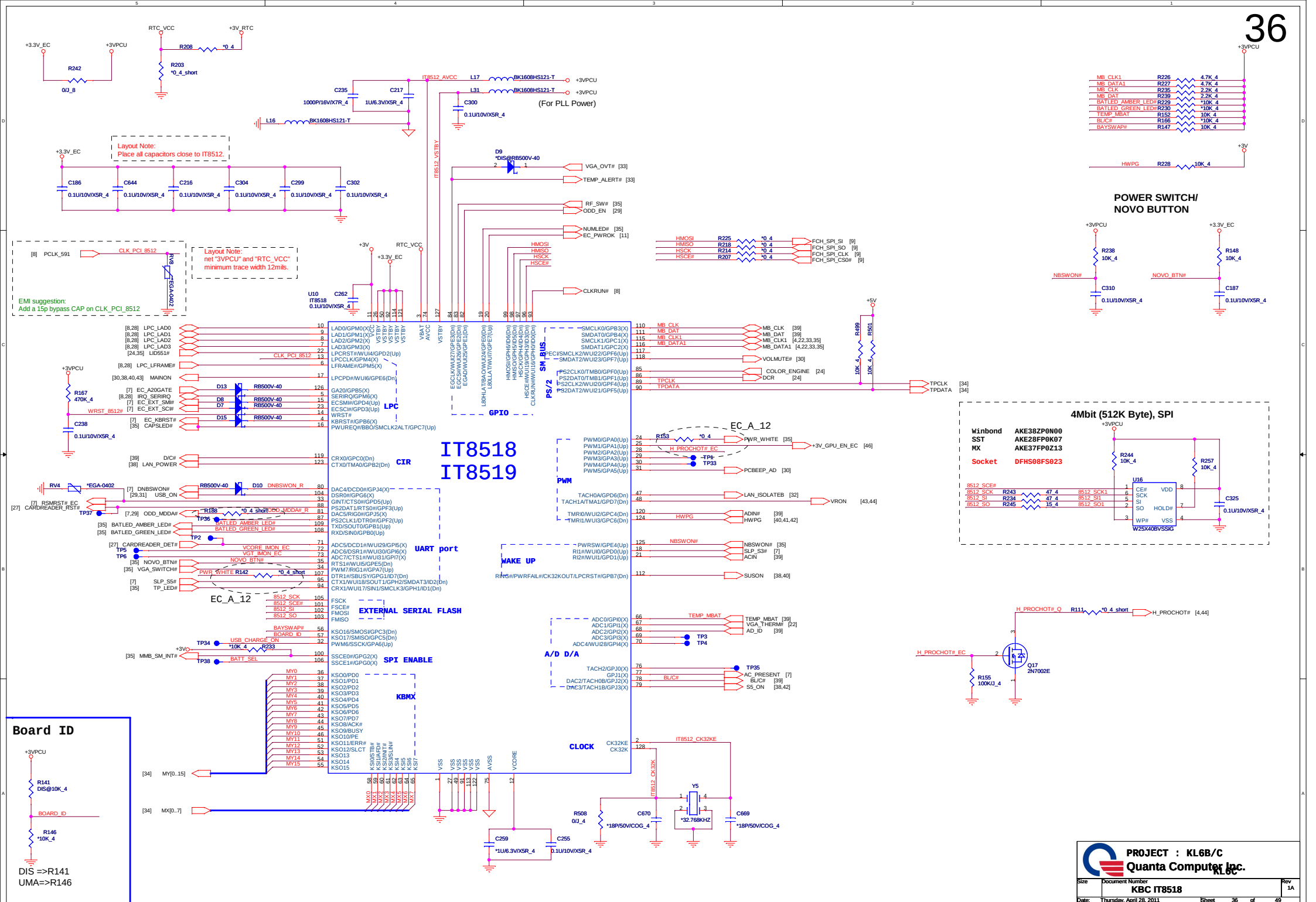
KEYBOARD



Touch pad





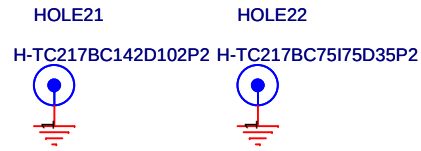


Screw for ME

CPU BKT

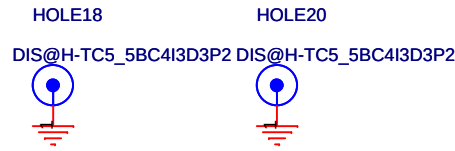
37

WLAN

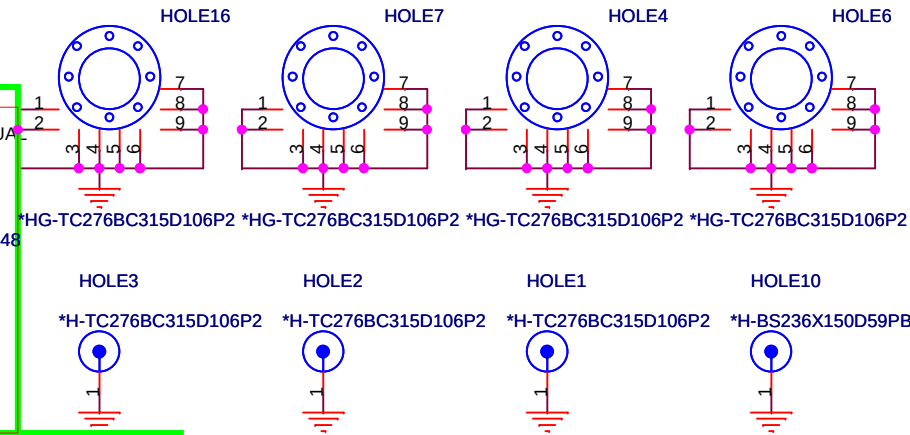
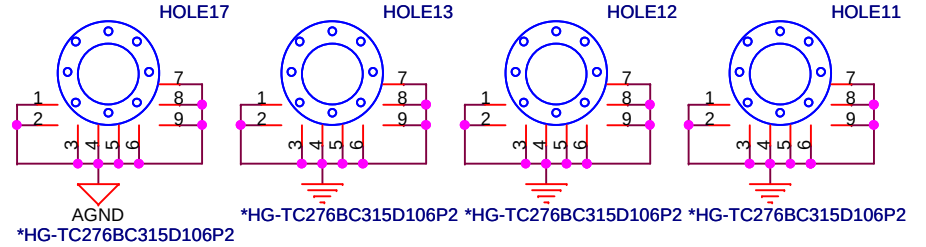
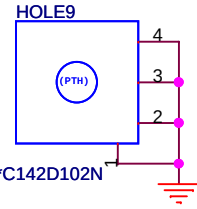


SMT NUT H=4/7mm

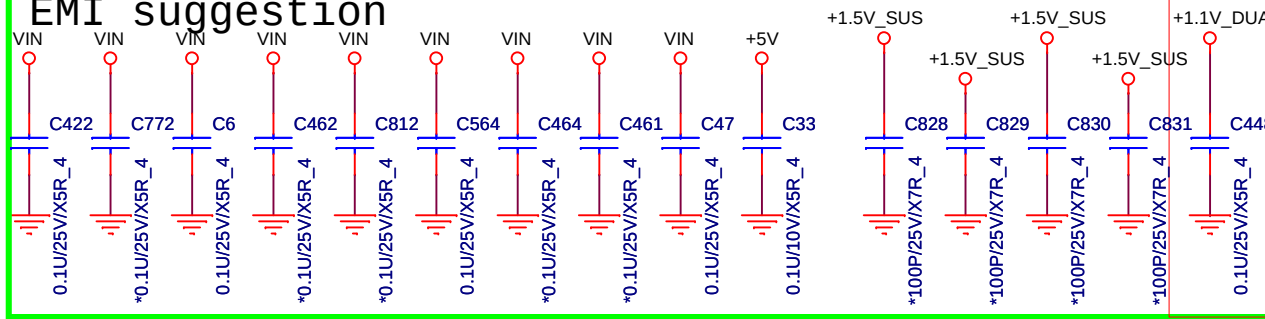
VGA



SMT NUT H=4/7mm



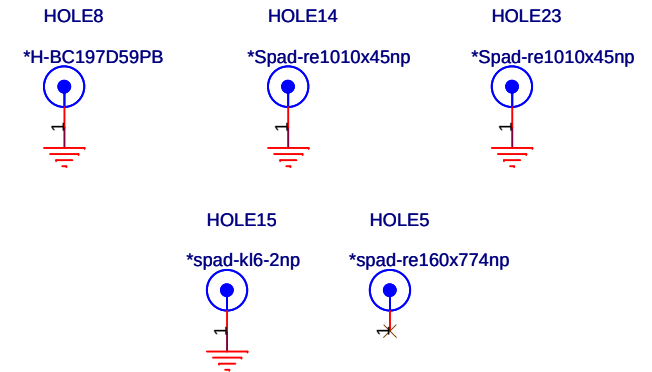
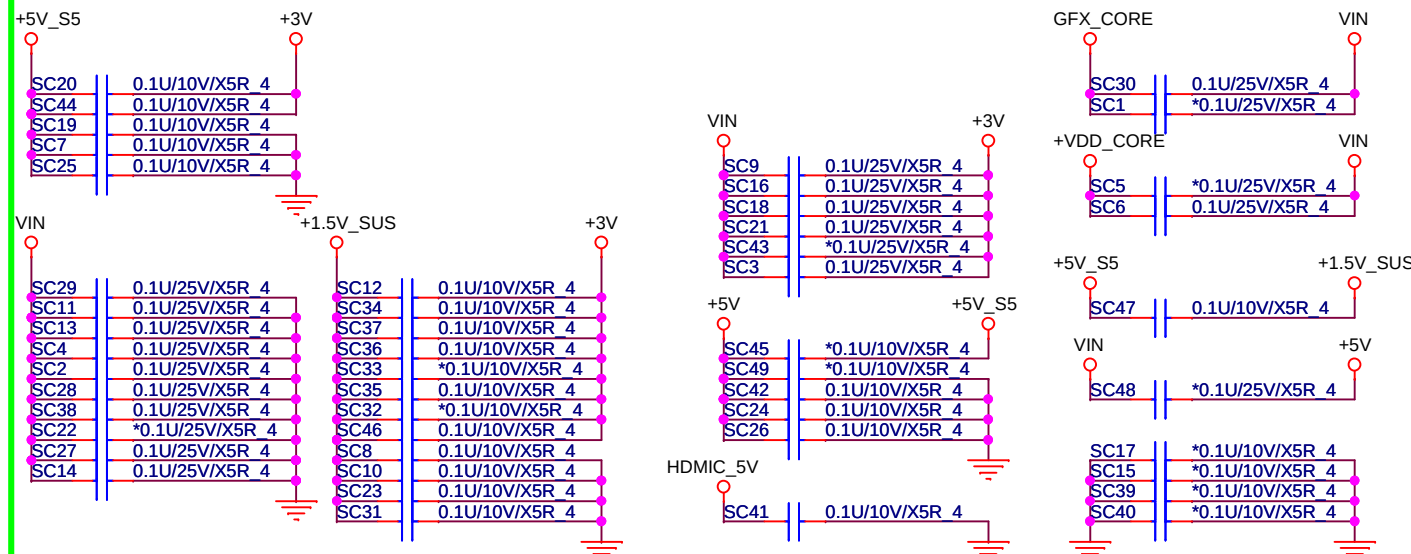
EMI suggestion



B-08

4/22 change

ESD suggestion

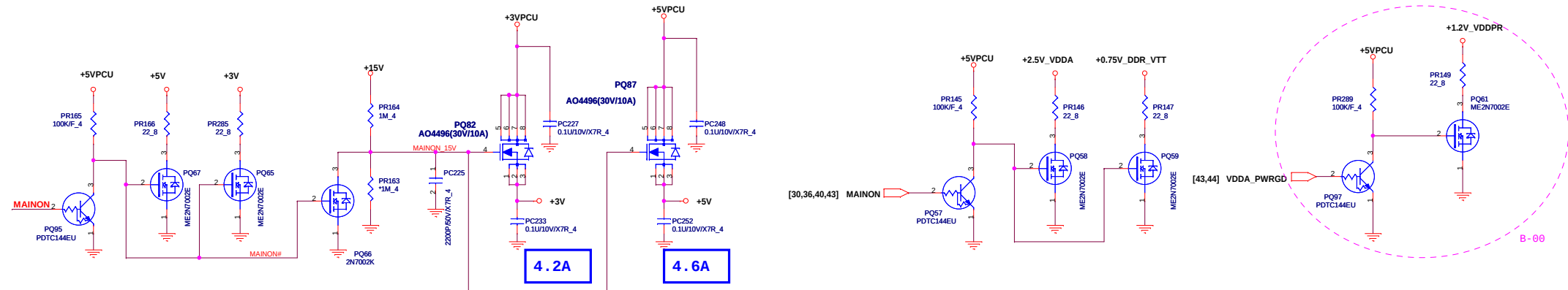
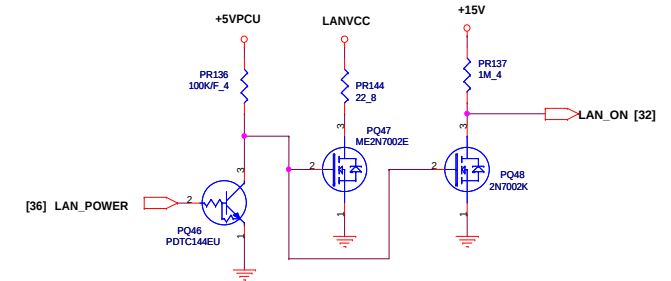


PROJECT : KL6B/C
Quanta Computer Inc.

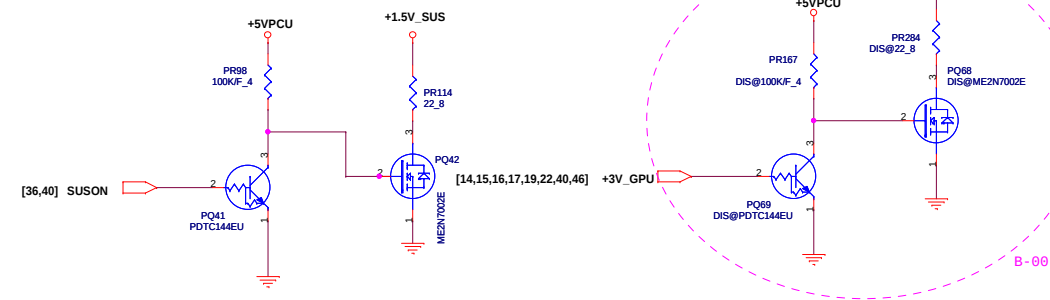
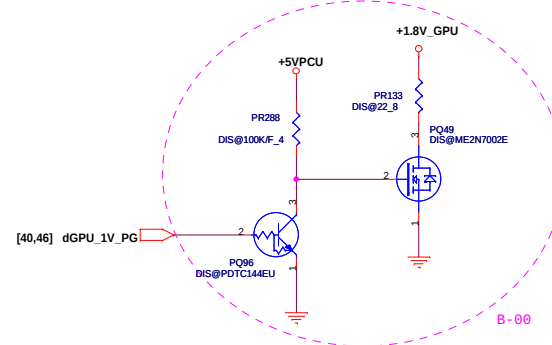
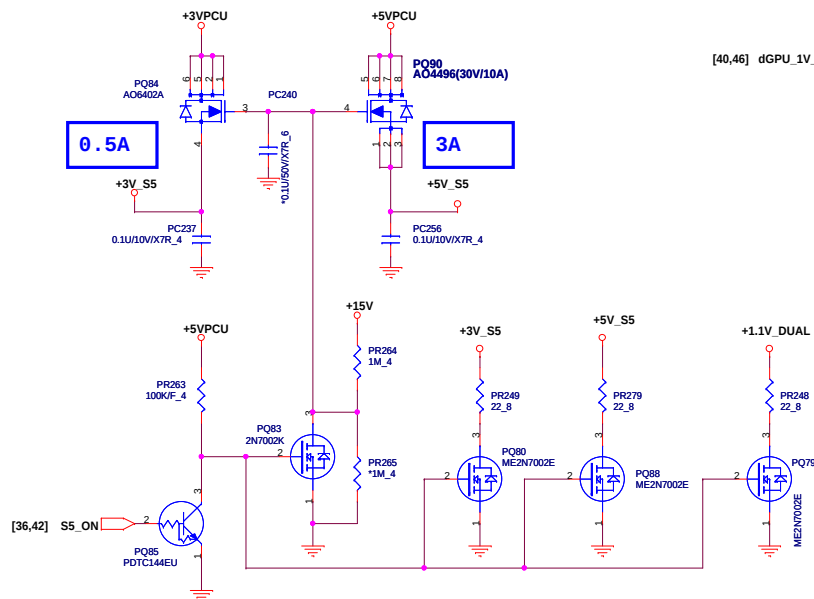
Size Custom	Document Number HOLD & SKEW	Rev 1A
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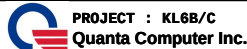
DISCHARGE

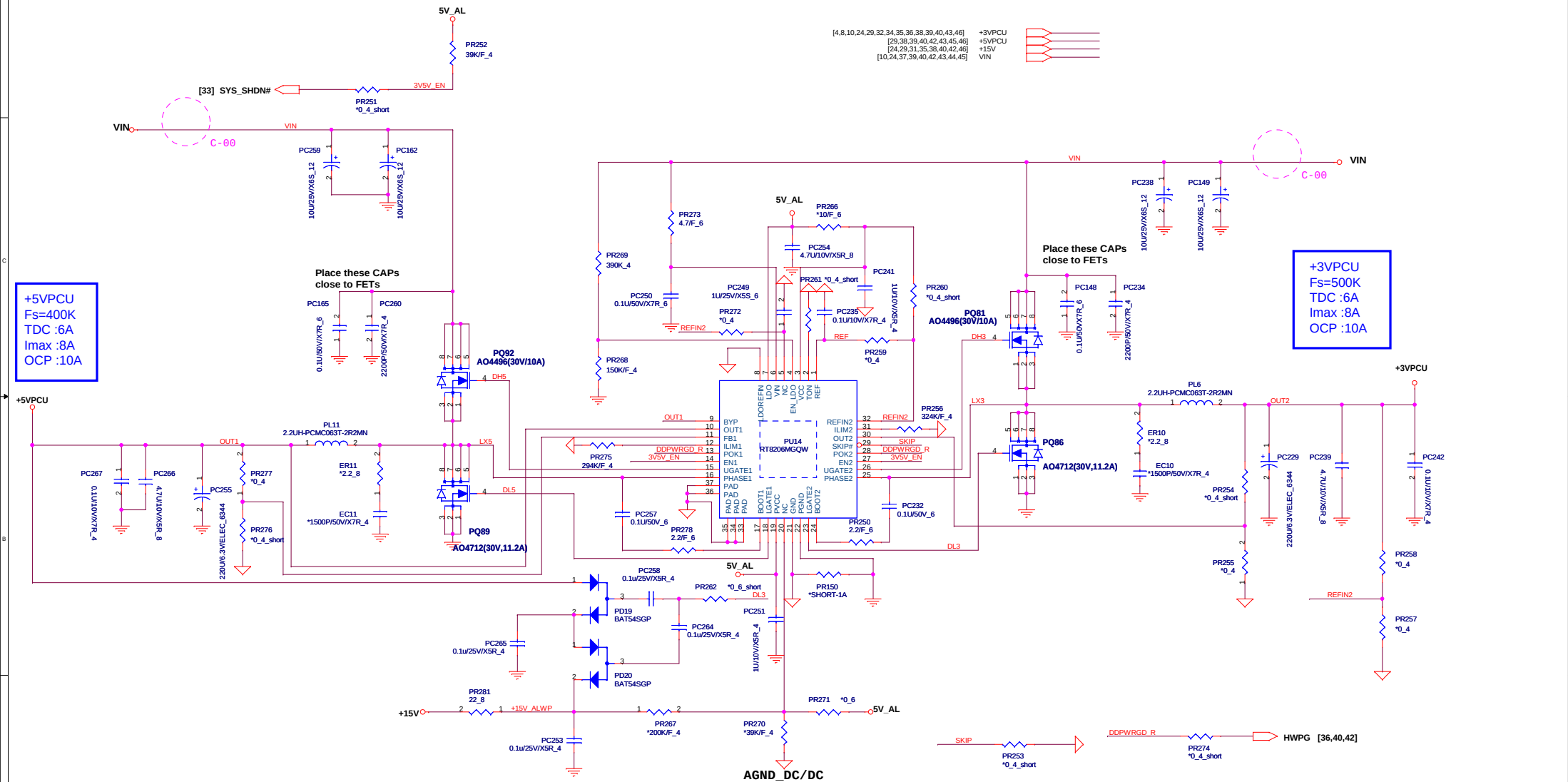
+3V, +5V, +1.5V

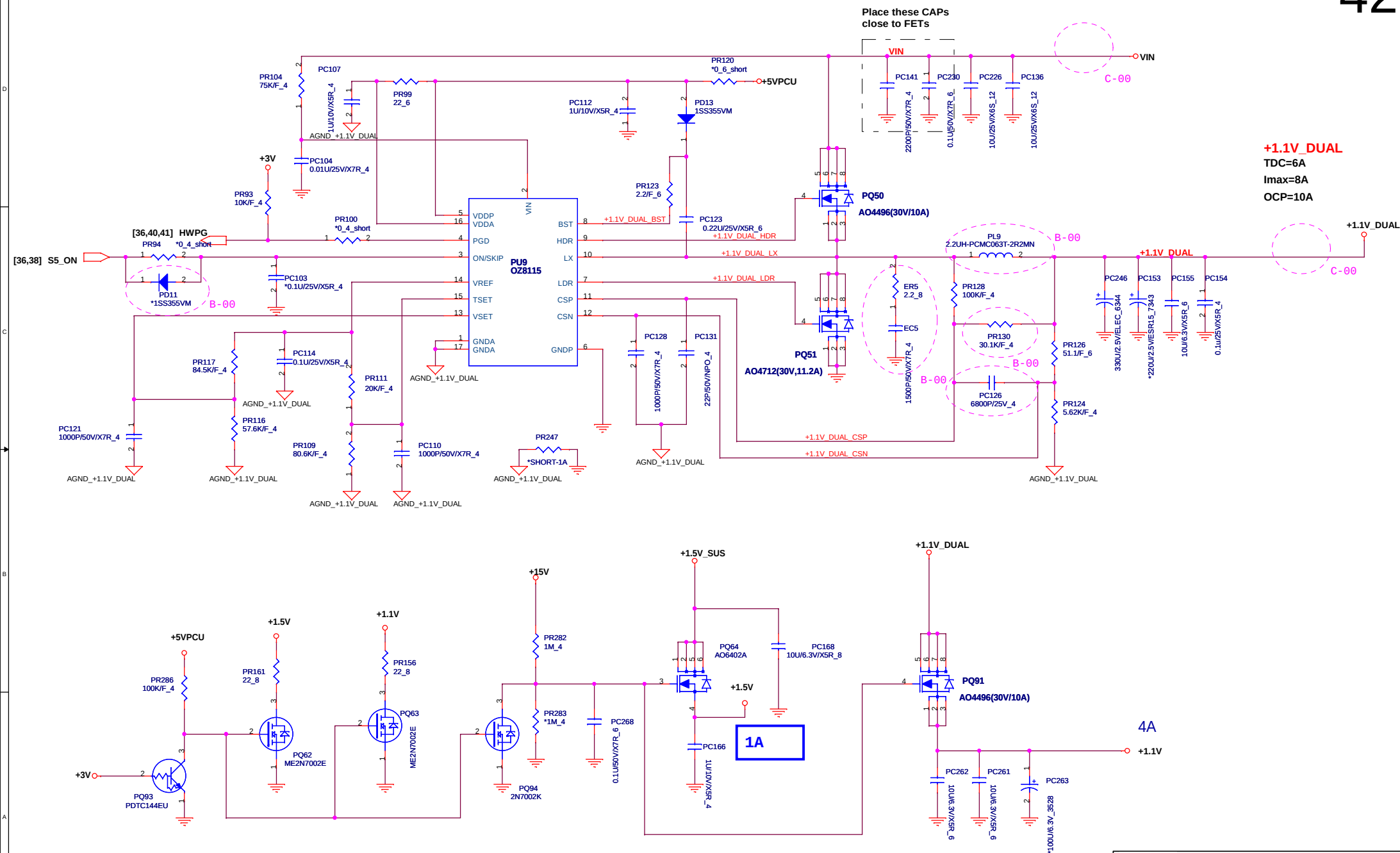
**LANVCC**

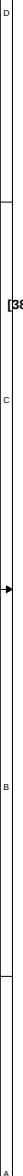
3V_S5, 5V_S5

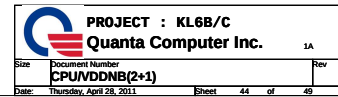


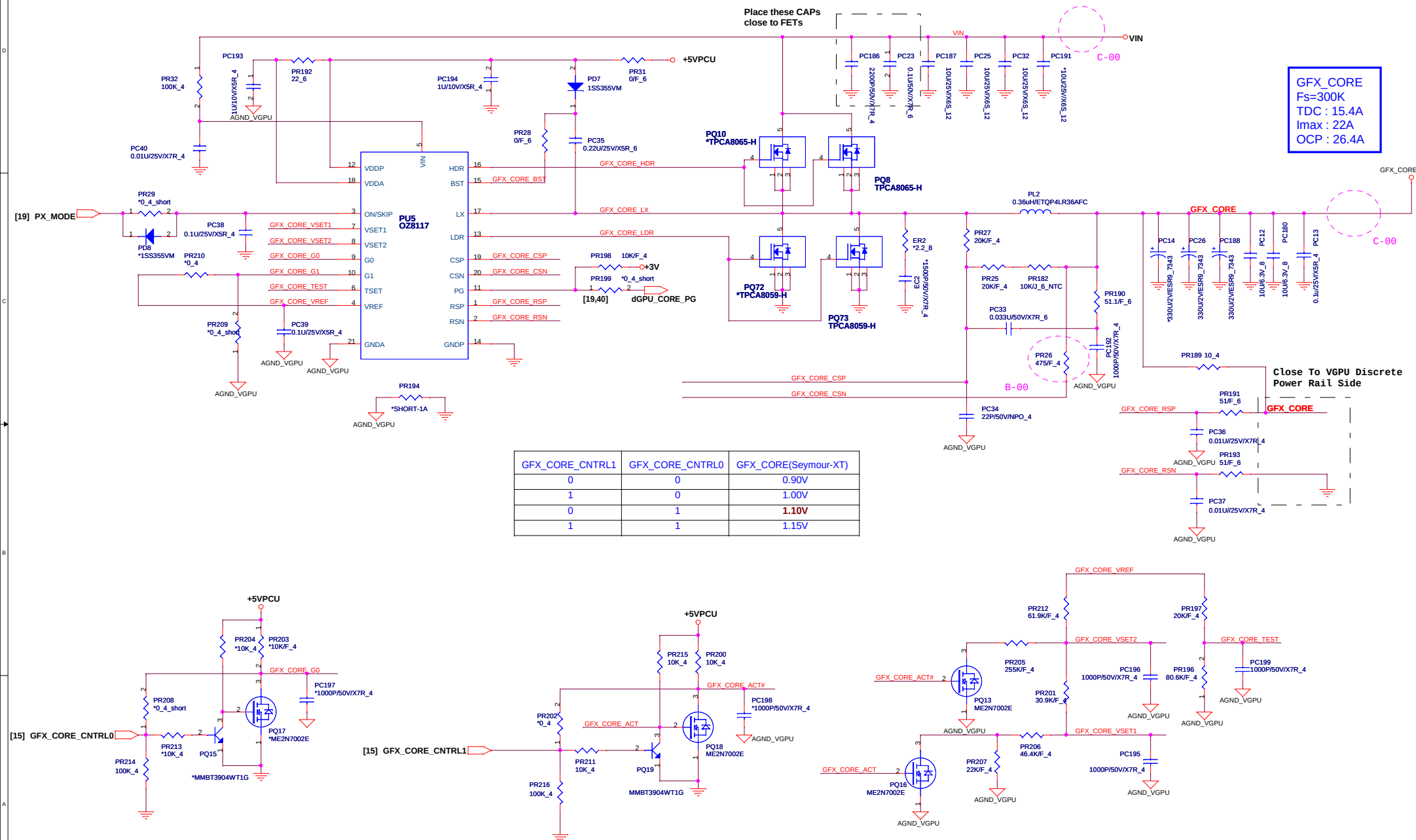


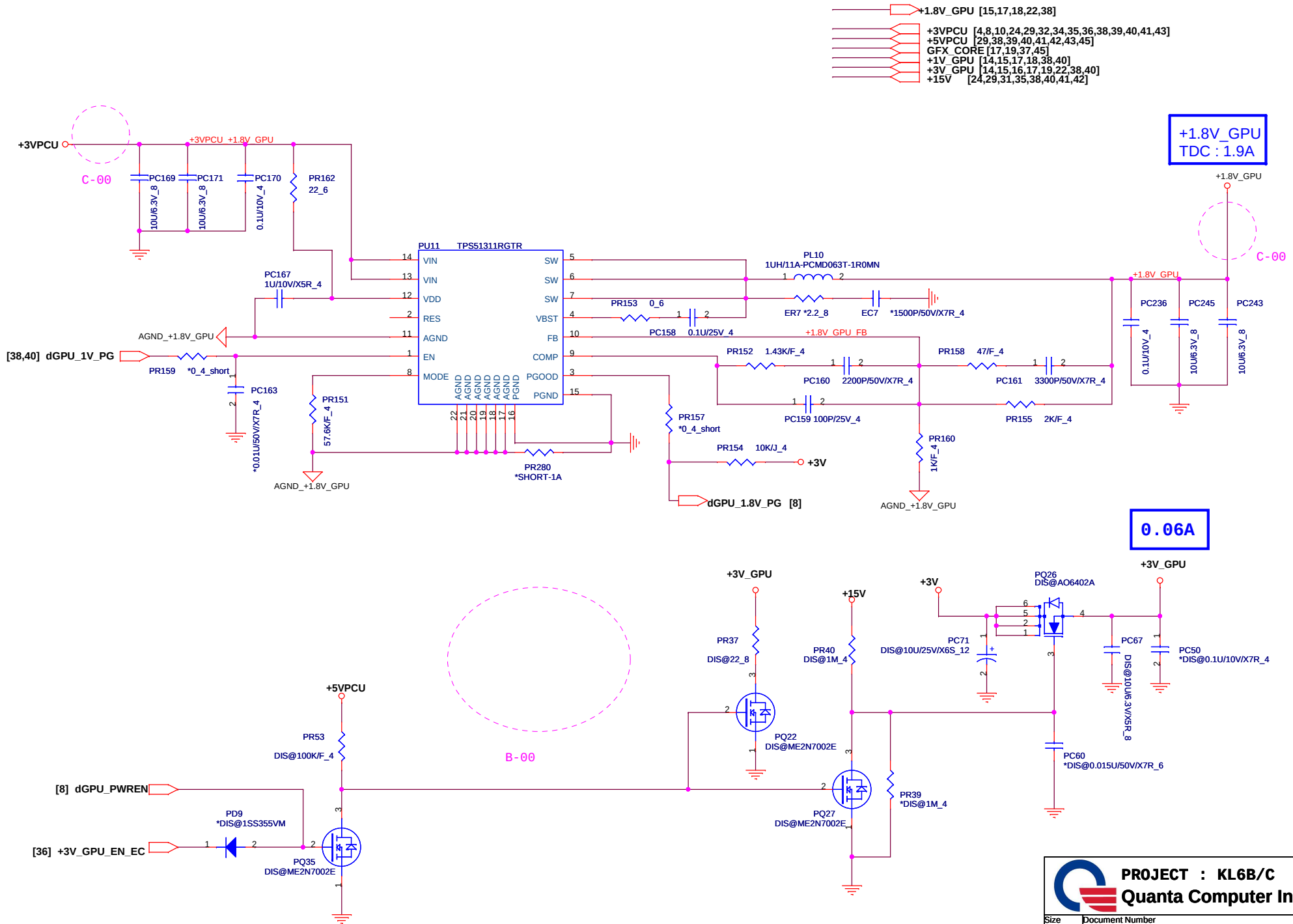




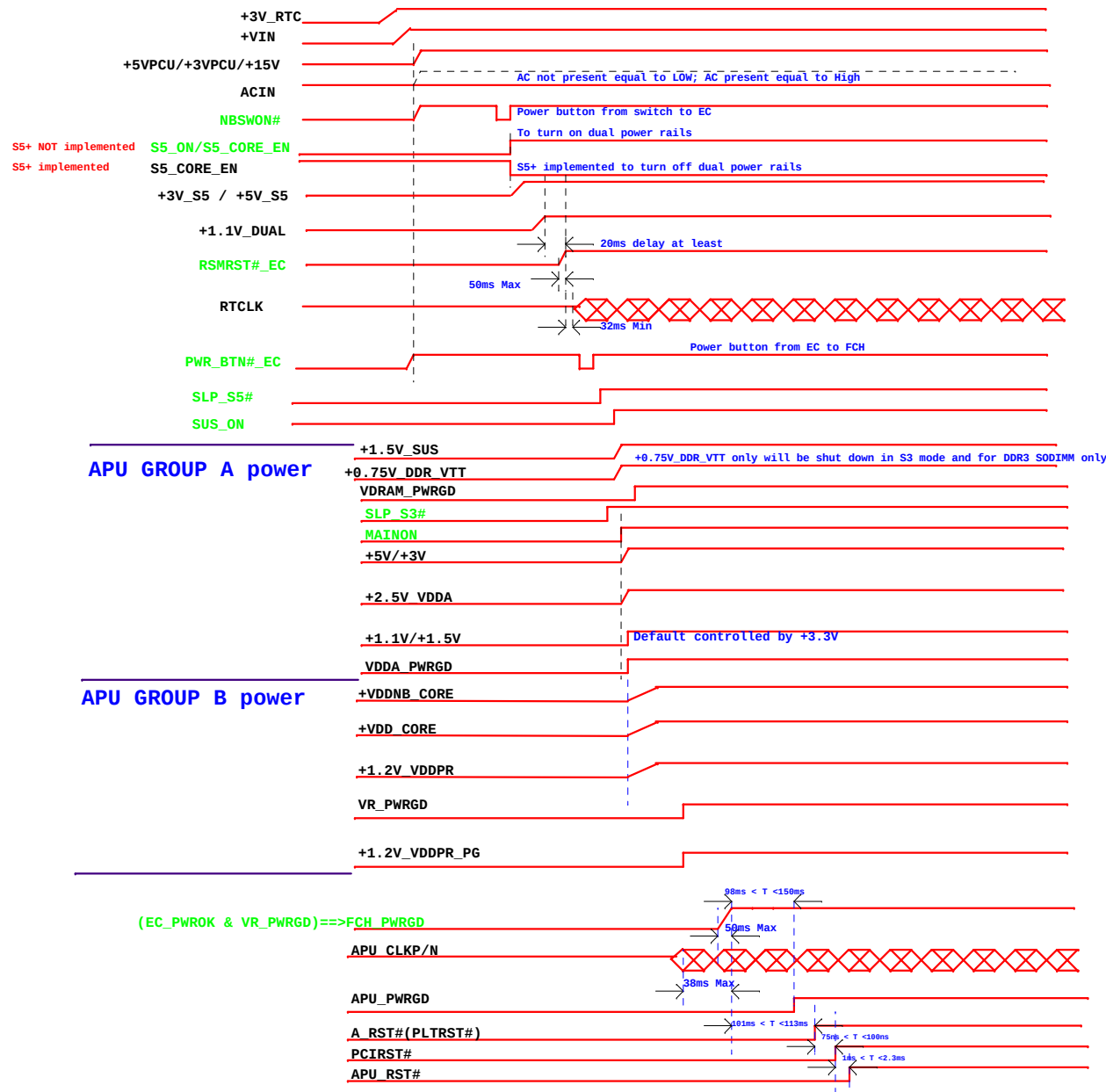








Z475 Power On Sequence: S5 > S0



APU Power on sequence required:

Llano APU:

- 1.Group A (+1.5V_SUS, +2.5V_VDDA) ramp before Group B (+VDD_CORE, +VDDNB_CORE, +1.2V_VDDPR)

HUDSON-M2/M3:

- 1.+3V_S5 ramp before +1.1V_DUAL
- 2.+3V ramp before +1.1V
- 3.+3V_RTC must ramp at least 5 secs before the +3V_S5

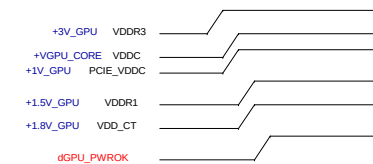
Robson-Pro M2/Capilano-LP M2 package Power-on sequence

All power rails reach nominal within 20ms

- 1=> +3V_GPU
- 2=> +VGPU_CORE/+1V_GPU
- 3=> +VGPU_CORE PWRGD to enable +1.5V_GPU
- 4=> +1V_GPU PWRGD to enable +1.8V_GPU

NOTE

- 1.EC (EC_PWR_EN) or FCH (PE_GPI01) to turn on +3V_GPU
- 2.+3V_GPU ready to enable +VGPU_CORE/+1V_GPU (+1V_GPU will ramp up before +VGPU_CORE)
- 3.+VGPU_CORE PWRGD to enable +1.5V_GPU
- 3.+1V_GPU PWRGD to enable +1.8V_GPU



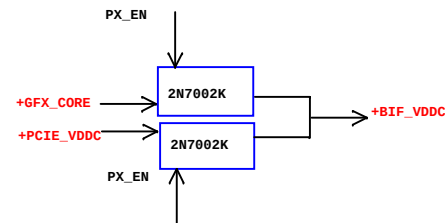
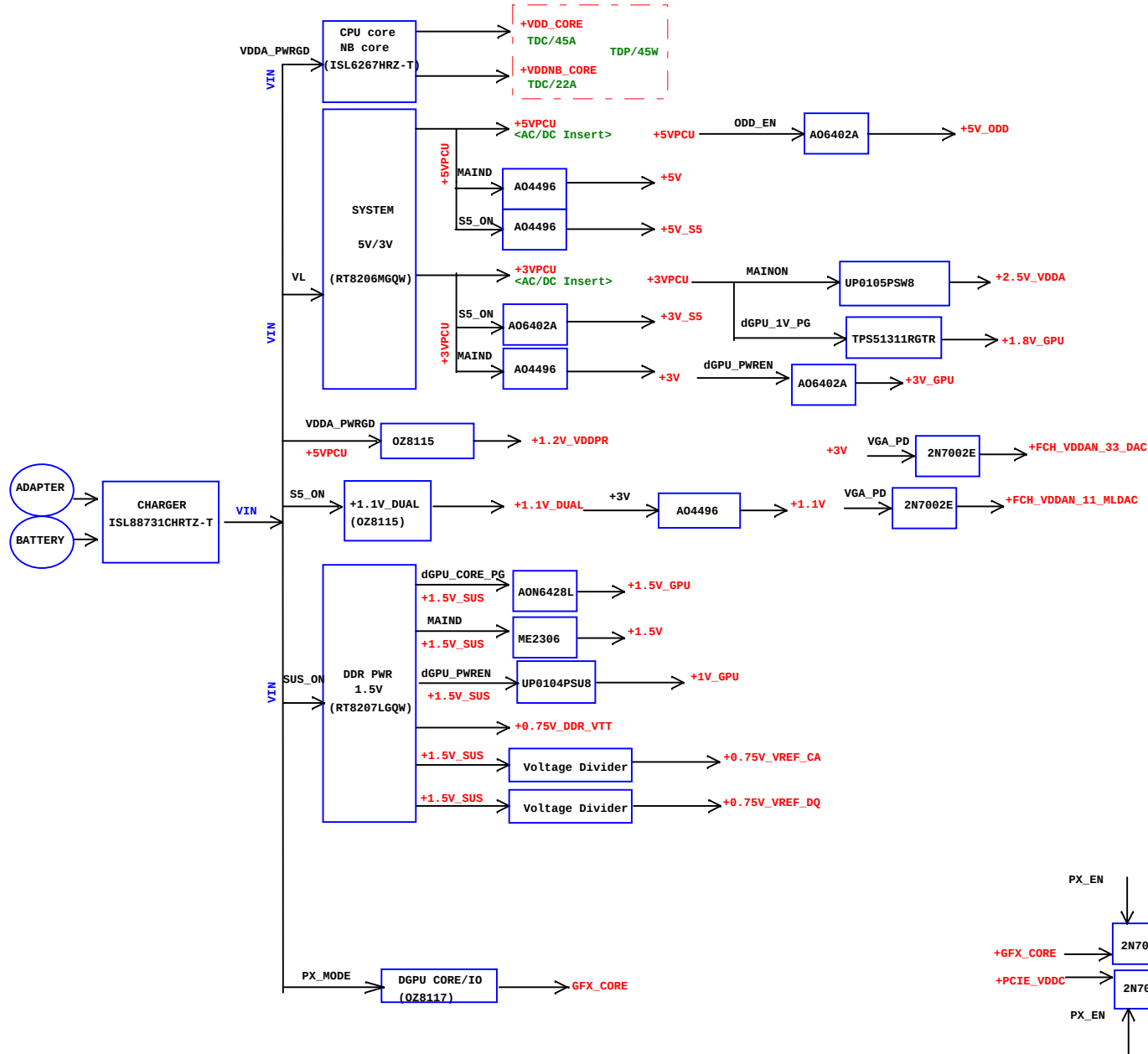
TRAVIS_L ANX3110 power sequence

- 1.+3V must lead +1.2V_TRAVIS

NOTE: EC output TRAVIS_EN# after receive +1.2V_VDDPR_PG

- 2.+1.2V_TRAVIS must lead TRAVIS_RST#

NOTE: FCH must output PCIE_RST#_TRAVIS or APU_PCIE_RST# after +1.2V_TRAVIS ready



KL6C / Z475 Schematic EE EC Tracking list

EC #	Page	CMVC #	Description	Date	Part Affected
A-01	4		Add level shifter design for LVDS PWM signal	0208	
A-02	4		Sys_pwrok pull high from 10K to 100K	0208	
A-03	9		Design issue need to swap SPI SI/SO signal	0208	
A-04	11		Disable Strap pin (EC disable) for CRT no function issue	0208	
A-05	14		DGPU reset change design to be controlled by GPIO only	0208	
A-06	16		Disable DGPU JTAG DEBUG MODE	0208	
A-07	23		Remove Travis IC power switch schematic for cost down	0208	
A-08	23		Design issue need to swap LVDS EDID signal	0208	
A-09	35		Change Resistor vaule to adjust the brightness of LED	0208	
A-10	7		Change USB port for USB2.0 & CardReader to USB.(Disable Hub3)	0208	
A-11	25		Change PU resistor value from 100ohm to 100kohm	0208	
A-12	36		Change EC pin to default PL to avoid PWR LED light one time as DC-IN	0208	
A-13	25		I2C level shift IC change to stuff.	0211	
A-14	22		no need to enable HDMI audio function in DGPU	0216	
A-15	7		Add Thermal trip PU resistor	0216	
A-16	18		Remove bead & cap for cost down	0218	
A-17	35		Add R188 ODD_MDDA# to connect to EC	0224	
A-18	35		Remove VGA SW2(ATI can't support HW switchch)	0224	
A-19	22		Remove VGA External thermal IC	0224	
SIV					
B-01	35		Add CCD P_SW on/off (Add C578, Q62, R467, Q63, R468 ; Del R455)	0306	
B-02	33		Connect to VGA Therm_Die use Channel 1.Del Q58(WLAN) to for CPU used. Add: R78, R77 ; Del: Q58	0306	
B-03	9		Change U17 VDD to +3V_S5	0306	
B-04	9		ADD SPI_HOLD# GPIO control, reserve R469	0306	
B-05	4		ADD SM_LV_CLK & SM_LV_DAT to connect FCH; Reserve R496 & R497	0306	
B-06	9		AUXCAL change to power rail for VGA Power Down Feature	0306	
B-07	22		ADD MOSFET for VGA SMBUS; ADD Q19, Q20, R158, R159	0306	
B-08	23		Del R8 ,R177,R99, R41,R42,R43 for remove LVDS reserve schematic	0307	
B-09	25		Delete R389,R401,D20,D21,D22,D23,Q36,Q37 cost reduce fail	0316	
B-10	25		Add R657,R658,R659,R660,R661,R662,R663,R664,C826,C827,U49	0316	
			2nd source verify	0316	
B-11			EMI solution	0316	
			POP R602,R601,R598,R597 120ohm	0316	
			POP CML3,CML4,CML2,CML1,L47	0316	
			Remove R464,R456,R599,R600,R451,R453,R412,R413,R465,R466 0ohm	0316	
			POP C408,C422,C749	0316	
			POP C391,C392	0316	
			Reserve C828,C829,C830,C831	0316	
B-12	06		POP R655,R656 Remove U18,R337,R330 Internal notice update	0317	
			Change Q61 PN:BA05L060000 Internal notice update	0317	
			Remove C207	0317	
B-13	34		CN5 Modify footprint to 85208-24071-24p-l	0317	
B-13	31		Add C832 for EMI solution	0318	
B-14			Change C515 & C516 to 27pF to fine tune timing	0321	
			Change C387 & C386 to 27pF to fine tune timing	0321	
			Change C188 & C184 to 27pF to fine tune timing	0321	
			Change C365 & C366 to 27pF to fine tune timing	0321	
SVT					
C-01	04		Change R541,R547 to *0_4_short	0418	
	06		Change R535,R537,R545 to *0_4_short	0418	
			Delete U18,R655,R656,R337,R330; J1,R587,R336,R323,R305,R580,R577	0418	
	07		Change R468,R287,R545,R169, R521,R533,R256,R317,R258,R552,R528 to *0_4_short	0418	
	08		Change R200,R183 to *0_4_short; Change R427 to *0_6_short	0418	

Document Number
EC Track

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Size Custom	Document Number EC Tracking Record A	Rev 1A
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KL6C / Z475 Schematic EE EC Tracking list					
EC #	Page	CMVC #	Description	Date	Part Affected
C-01	09		Change R263,R254,R247 to *0_4_short	0418	
	19		Change R157 to *0_4_short	0418	
	23		Change R83,R50,R46,R91 to *0_4_short; change R650 to *0_8_short	0418	
	24		Change R2 to *0_4_short; Change R7,R28 to *0_8_short	0418	
	25		Change R660,R661,R662,R663 to *0_4_short	0418	
			Delete U42,R603,R605, R381,R382,R384,R386,R399,R400; C375, C377,C448,C449,C459	0418	
	28		Change R398,R424,R428 to *0_4_short	0418	
			Change R397 to *0_8_short	0418	
	29		Change CML1 footprint to "choke-wcm2012-4p"	0418	
			Delete R412,R413; Change R365,R3698,R604 to *0_8_short	0418	
	31		Change R438 to *0_4_short	0418	
			Change CML2,CML3,CML4 footprint to "choke-wcm2012-4p"	0418	
			Delete R451,R453,R599,R600,R456,R464	0418	
	32		Change R444 to *0_4_short; Change R418,R421,R435,R449 to *0_8_short	0418	
	33		Change R77,R78 to *0_4_short; Change R595 to *0_8_short	0418	
	34		Change L34 to *0_8_short	0418	
	35		Change L47 footprint to "choke-wcm2012-4p"	0418	
			Delete R465,R466; Change R161,R162 to *0_6_short	0418	
	36		Change R111,R142,R188,R203 to *0_4_short	0418	
	27		Change R646 to *0_8_short	0418	
	12		Change R394 to *0_6_short	0418	
C-02	31		change U47 Pin8 design	0422	
	37		Add C448 (0.1u CAP) for EMI solution	0422	
C-03	4		delete APU debug funciton, R570, R579,R583,R584,R596	0425	
	34		Mount CA1,CA2,CA3,CA4,CA5,CA6 for EMI solution	0426	
	35		Change Q62 PN to BAM34040001	0426	
	10		Change Q29,Q32 PN to BAM34040001	0426	
C-04	25		Change value of R601,R602,R597,R598 to 140 ohm for HDMI issue	0428	
			Change value of R383,R385,R387,R388,R391,R392,R395,R402 to 560 ohm for HDMI issue	0428	